

Cape Town | Report

2025

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0. Abstract

Cape Town stands at a critical juncture in its urban evolution, confronting a spatial legacy that continues to impose profound economic and social costs on its most vulnerable residents.

Despite the political transformation that began in 1994, Cape Town remains South Africa's most segregated city, with social inequalities continuing to align with race even as the underlying causes have shifted from apartheid-era controls to economic and labour-market dynamics¹. The city's spatial form, characterised by a stark divide between well-resourced economic nodes and marginalised townships on the periphery, creates a daily crisis of access affecting millions of residents who must navigate vast distances between home and opportunity.

This Report presents the Opportunity Hub Network as a strategic, evidence-based intervention. Building on comprehensive spatial analysis that identified critical deficits in public space provision, transport connectivity, and access to economic opportunities, this report develops a networked solution that operates at both micro (site-specific) and macro (city-wide) scales. The research responds to dual commissions from the City of Cape Town's Department of Urban Design and the Mission for Inner City Cape Town, providing both metropolitan-scale analysis of public space provision and actionable implementation strategies for immediate intervention.

¹ Turok et al., 2021

1. Summary



Cape Town photographed from Lions Head
Source © [Cape Town] Cohort, Norman Foster Institute. 2025

1.1 Statement of Purpose

This report aims to demonstrate how Cape Town can transform its transport interchanges from spaces of waiting into platforms for opportunity. It presents a data-driven, design-led methodology for addressing spatial inequality through the deployment of modular Opportunity Hubs at priority nodes. By combining rigorous analysis, community co-design, and scalable construction systems, the report provides a practical roadmap for implementing catalytic interventions that activate public space, expand access to services, and lay the foundation for permanent urban integration.

1.2 Scale of the Challenge

The urgency of intervention cannot be overstated. Currently, 89% of commuter trips in Cape Town are road-based, with 57.5% of commuters using private vehicles while only 1.3% use passenger rail due to system collapse². Cape Town drivers spent an average of 94 hours stuck in traffic in 2024, marking the city as the most gridlocked in South Africa and representing a 13% increase from the previous year³. For township residents dependent on public transport, these systemic failures translate into daily journeys that can exceed four hours and consume up to 40% of household income, a burden that perpetuates cycles of poverty and exclusion.

Beyond the journey itself, hundreds of thousands of commuters spend additional hours each day waiting at transport interchanges in conditions that lack basic dignity: places without seating, shade, productive activities, or services. At Cape Town Station Deck alone, over 200,000 daily commuters navigate hostile public spaces while waiting. These nodes, where people are forced to dwell rather than simply pass through, represent both the daily indignity of spatial apartheid and the immediate opportunity for transformation. The Opportunity Hub Network recognises that these transport interchanges constitute underutilised urban assets where strategic intervention can reach the maximum number of people without requiring additional travel, the foundational principle of the last-mile approach that structures this methodology.

² City of Cape Town, 2024

³ INRIX, 2024



Cape Town from above showing the Sprawl of the City
Source © [Cape Town] Cohort, Norman Foster Institute. 2025

1.3 A Network Approach

The Opportunity Hub Network represents a departure from traditional infrastructure-heavy approaches to spatial transformation. Instead of waiting for large-scale transport upgrades or massive housing developments, the intervention deploys rapidly implementable, cost-effective solutions that can begin delivering benefits within 12 months. The 2026 pilots in Langa and the Central Business District, including spaces around Station Deck, will demonstrate how strategically placing basic services, workspaces, skills programmes, enterprise support, and e-services can create local opportunities, and transform hostile public spaces into productive community assets.

What distinguishes this intervention from previous attempts at improving spatial equity in Cape Town is its radical integration of functions that are traditionally delivered through separate programmes and departments. Each hub combines workspace provision addressing the lack of productive spaces in townships, skills development aligned to actual employer commitments, enterprise support spanning registration to procurement linkages, public e-services that reduce trips for government transactions, basic service provision to improve access to essential utilities, and public realm upgrades that create safe and dignified public spaces.

This integration reflects international best practices in place-based development while responding to Cape Town's specific context of mounting pressure from in-migration, expanding informal settlements, and overloaded infrastructure in newly urbanised zones⁴.

⁴ Gildenhuys, 2025



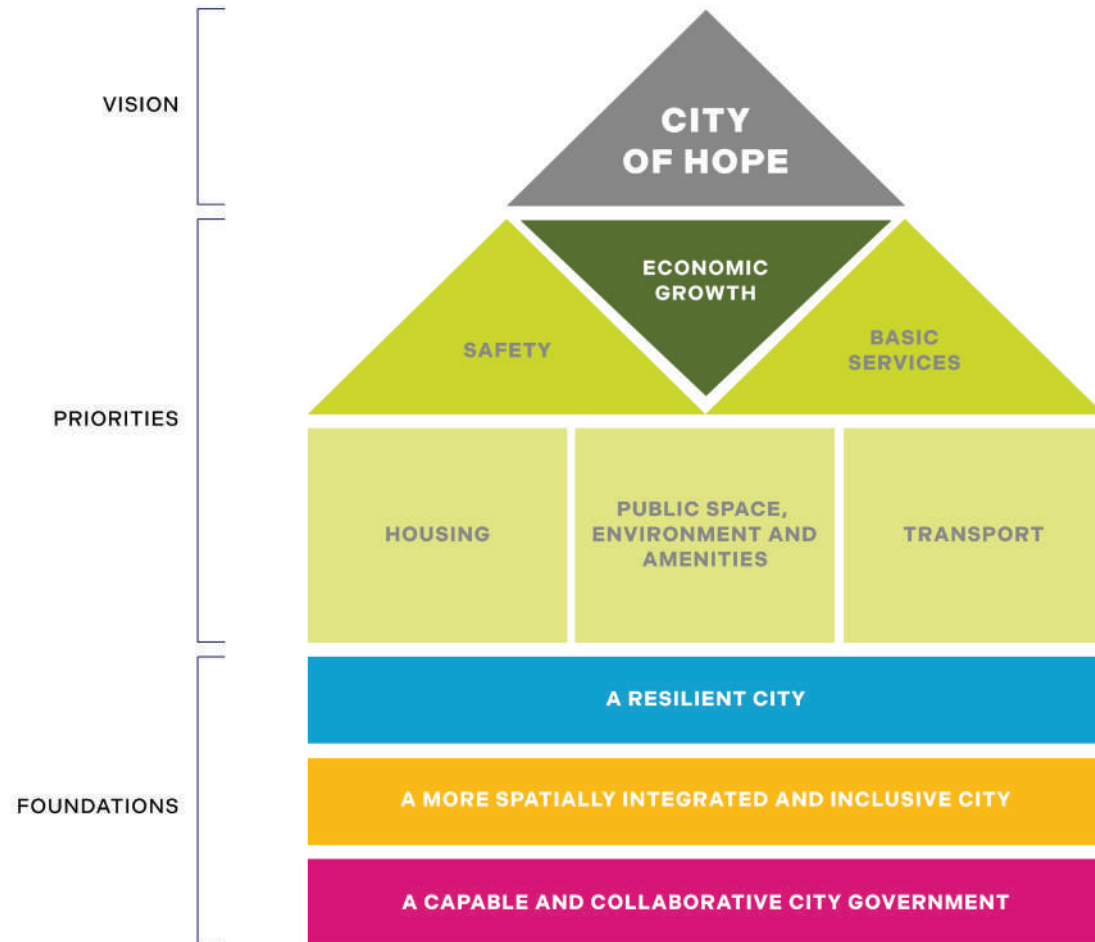
People waiting in Line at a Transport Hub in Cape Town
Source © Harrison, David. Mail & Guardian (online). 2021.

1.4 Policy Alignment

The intervention directly responds to the City of Cape Town's Integrated Development Plan (IDP) 2022-2027, which sets the city's strategic priorities for inclusive growth, service delivery, and spatial equity, and their Comprehensive Integrated Transport Plan (CITP) 2023-2028, which outlines a long-term vision for an efficient, safe, and accessible transport system guided by a triple access paradigm of mobility, proximity, and digital connectivity⁵.

The policy framework of the Integrated Development Plan provides explicit support for hub network objectives across multiple dimensions: Jobs and Investment (Objective 1) through workstations, connectivity, skills training, and enterprise support; Quality Public Places (Objective 11) through lighting, shade, seating, and dignified public realm upgrades; Safe and Affordable Transport (Objectives 12-13) through strategic placement at public transport interchanges; Spatial Integration (Objective 15) through development of clustered hubs into corridor networks and eventually township employment districts; and Support for Micro-Developers (Objective 8.1) through co-located advisory services supporting incremental densification near hubs.

By embedding modular, multi-functional hubs at strategic transport nodes, the network creates local pathways to employment, enterprise development, skills training, basic services and civic administration while simultaneously upgrading public realm conditions found in the city's Quality Public Places Programme standards.



⁵ City of Cape Town, 2022, 2023

Integrated Development Plan Diagram

Source © City of Cape Town. Integrated Development Plan 2022-2027.

1.5 Method Summary

This report introduces a five-stage methodology that bridges quantitative spatial analysis with participatory design and modular construction. Stage 1 employs the Opportunity Explorer, a site prioritisation tool analysing 610 transport nodes across five Key Performance Indicators: Basic Services, Learning and Skills, Work and Enterprise, Digital and Media Access, and Civic and Administration. Stage 2 applies the last-mile principle to select specific sites around priority nodes. Stage 3 utilises a Catalogue Matrix System crossing spatial dimensions (site sizes) with temporal dimensions (dwell times) to generate appropriate intervention types. Stage 4 transforms the technical catalogue into collaborative community co-design processes. Stage 5 implements interventions through modular construction using standardised structural frames that enable both immediate implementation and long-term adaptability.

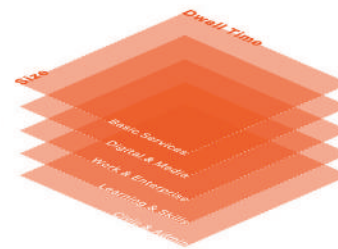
The methodology's true innovation lies not in any single component but in their systematic integration. Data analysis informs but does not determine design. Community knowledge supplements but does not override technical requirements. Standardised elements enable rather than constrain local adaptation. This careful calibration positions Opportunity Hubs as catalysts for broader transformation, creating the spatial and social infrastructure necessary for township communities to access opportunities that spatial apartheid continues to deny.



1 Prioritization



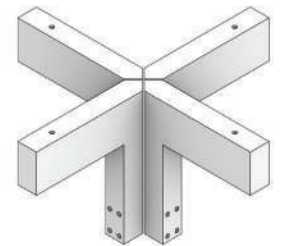
2 Site Selection



3 Catalogue



4 Participation



5 Intervention

2. Introduction

2.1 The Urban Crisis of Disconnection

Cape Town's contemporary urban form cannot be understood without acknowledging its deliberate construction through centuries of colonial planning and decades of apartheid spatial engineering. The city's urban evolution has been shaped by successive waves of colonial occupation, from Portuguese and Dutch settlements in the 15th and 17th centuries to British control in the 19th century, and by apartheid policies in the 20th century that formalised racial segregation through forced displacement and urban fragmentation.⁶ This historical trajectory has produced what urban geographers term a "dual city," in this case, where proximity to opportunity correlates directly with historical racial classification and contemporary economic status.

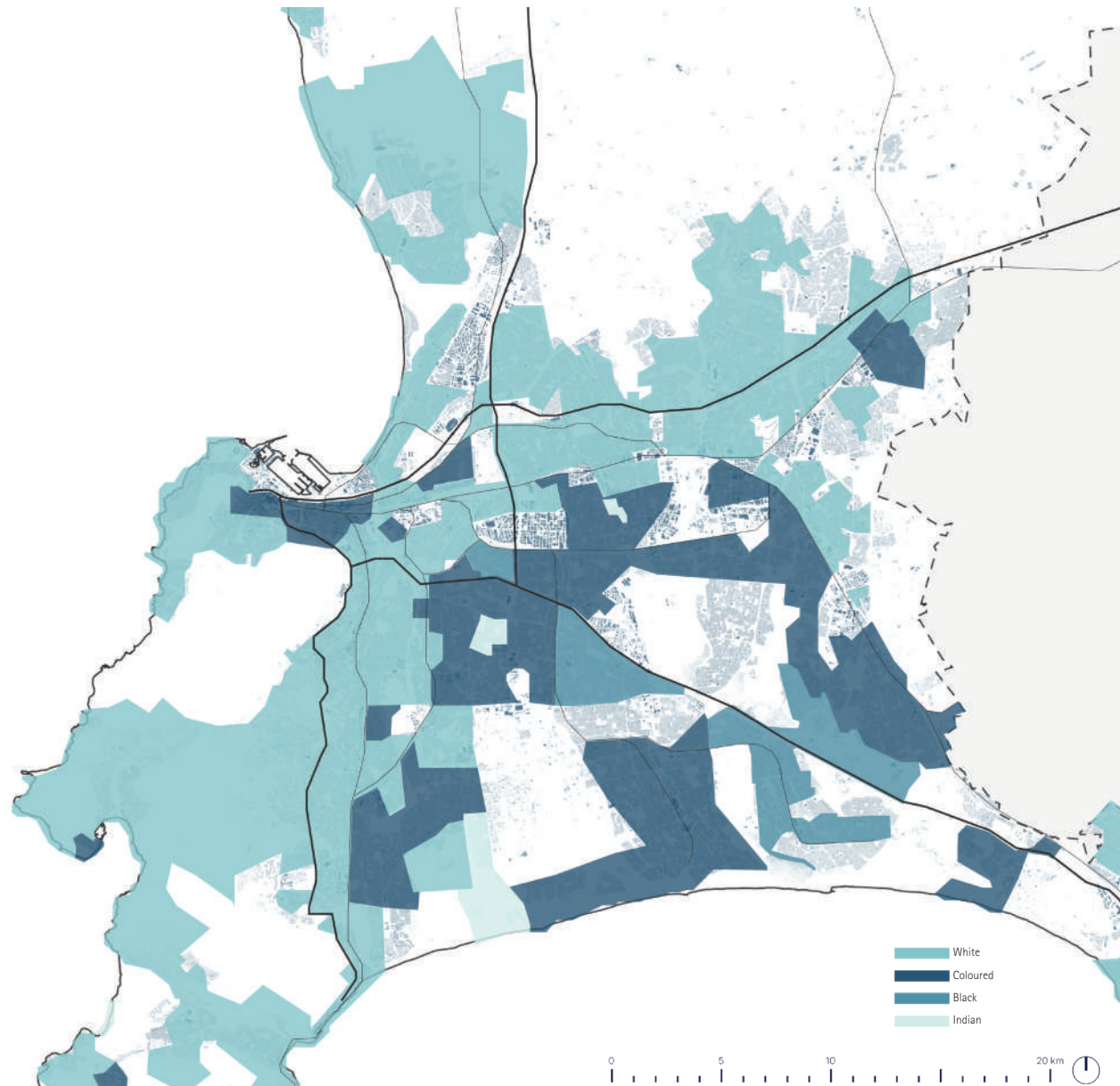
The persistence of this spatial inequality is evidenced in stark economic disparities. White residents earn, on average, three times more than Black, Asian, or Coloured populations, with many communities still facing barriers to basic services, education, and mobility.⁷ The City of Cape Town's Gini coefficient increased from 0.606 in 2012 to 0.626 in 2020, with income inequality levels generally higher across all racial groups compared to provincial averages⁸.

Three decades after the end of apartheid, Cape Town remains one of the world's most unequal cities. These disparities are not merely economic; they are fundamentally spatial. The physical structure of the city, shaped by decades of deliberate segregation, continues to determine who can access opportunity and who cannot.

⁶ Turok et al., 2021

⁷ Turok et al., 2021

⁸ Western Cape Government, 2023



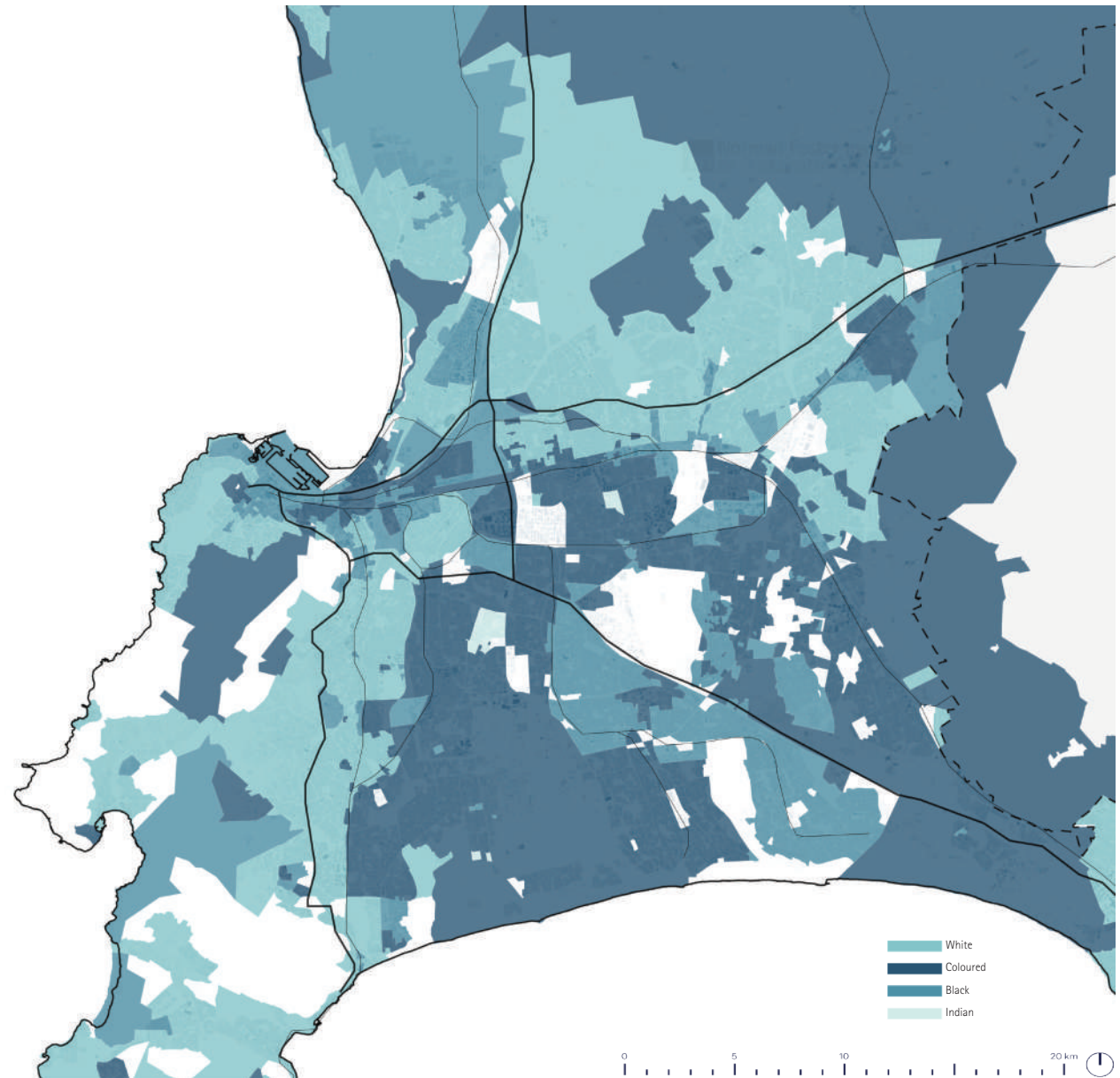
Group Areas Act (1948)

Sources © [Cape Town] Cohort, Norman Foster Institute. 2025
Crankshaw, Owen.

The apartheid regime's Group Areas Act assigned people to particular places kept apart by buffer strips, allocating central areas to white residents while confining Black and Coloured populations to peripheral townships on the Cape Flats.⁹ Implementation of this Act destroyed established communities, most notoriously District Six, from which approximately 55,000 residents were forcibly displaced. Though these laws were repealed in the 1990s, their spatial consequences endure. Roads like the N2 and the central railway line, originally designed to move low-wage labour from marginalized areas into the city centre, continue to structure movement and reinforce division.

The spatial dimension of inequality is captured by what UCT property economist Professor Francois Viruly, coined the "40x40x40x40 framework": township residents typically live in 40m² dwellings, located 40km from employment centres, spending 40% of household income on transport, in communities experiencing approximately 40% unemployment. While illustrative rather than statistically precise, this framework captures the compounding nature of spatial disadvantage. Households in townships experience longer commutes, fewer services, reduced access to employment, and diminished public space, demonstrating why individual interventions are insufficient to address entrenched inequality.

⁹ Turok & Scheba, 2019



Racial Distribution (2011)

Sources © [Cape Town] Cohort, Norman Foster Institute. 2025
City of Cape Town. Open Data Portal

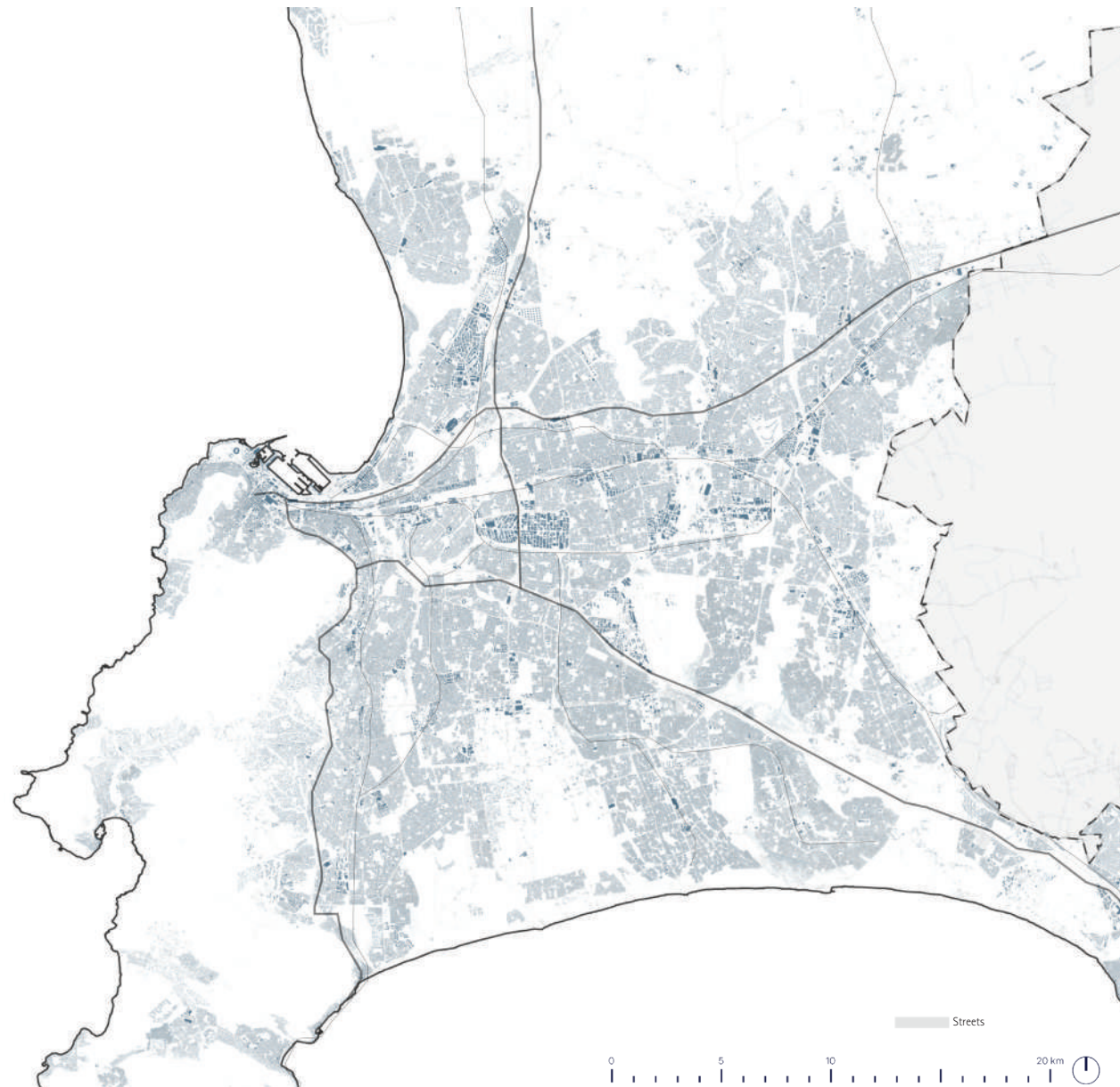
2.2 The Transport Burden as Structural Constraint

The daily commute from Cape Town's townships to employment centres constitutes more than an inconvenience; it represents what sociologist Johan Galtung would term "structural violence," in which social structures harm people by preventing them from meeting their basic needs. The busiest public transport interchanges, including Cape Town Station Deck (158,000 daily commuters), Bellville (108,000), and Khayelitsha (60,000), process hundreds of thousands of workers in conditions that often lack basic dignity.¹⁰ These are the locations where the Opportunity Hub Network concentrates its intervention, recognising that transport nodes represent the most strategic points for addressing spatial inequality precisely because there is a concentration of diverse populations and can transform unproductive waiting time into opportunities for economic and social engagement.

The collapse of the passenger rail system has intensified Cape Town's mobility crisis. Since 2019, weekday train services across the province have decreased by 66%, resulting in 293 fewer trains operating each day. Between 2012 and 2022, ridership on Cape Town's passenger rail network declined by 92% due to theft, vandalism, and poor management.¹¹ This decline forced commuters onto an already strained road network and pushed many toward minibus taxis, which have become the default public transport option for low-income residents despite being more expensive, less regulated, and unsafe. Minibus taxis, which carry 22% of daily commuters, operate with limited regulation and frequent safety incidents, yet remain essential due to their affordability and route flexibility.

¹⁰ City of Cape Town, 2024

¹¹ PRASA, 2021



Street Network

Sources © [Insert City Name] Cohort, Norman Foster Institute. 2025
City of Cape Town. Open Data Portal

The separation between where people live and where jobs exist imposes severe costs on township residents. The City of Cape Town's Comprehensive Integrated Transport Plan (CITP) documents that public transport users in the lowest income bracket spend an average of 17% of their monthly income on direct transport costs alone.¹² When indirect costs, including waiting time and total travel time, are included, the burden rises substantially. Research indicates that transport costs can consume up to 40% of low-income household budgets, leaving insufficient resources for food, education, and other essentials¹³.

Travel times compound these costs. According to the CITP's Urban Development Index, minibus taxi users average 39 min. of travel during morning peak periods, while bus users average 79 min¹⁴. These figures understate total journey times, as they exclude walking to stops, waiting, and transfers. For many households, journey times can exceed four hours daily, fundamentally constraining access to employment and reducing the effective labour market available to township residents. The average travel distance to work for public transport users is 17 kilometres, more than double the 8 kilometre average for private vehicle users.

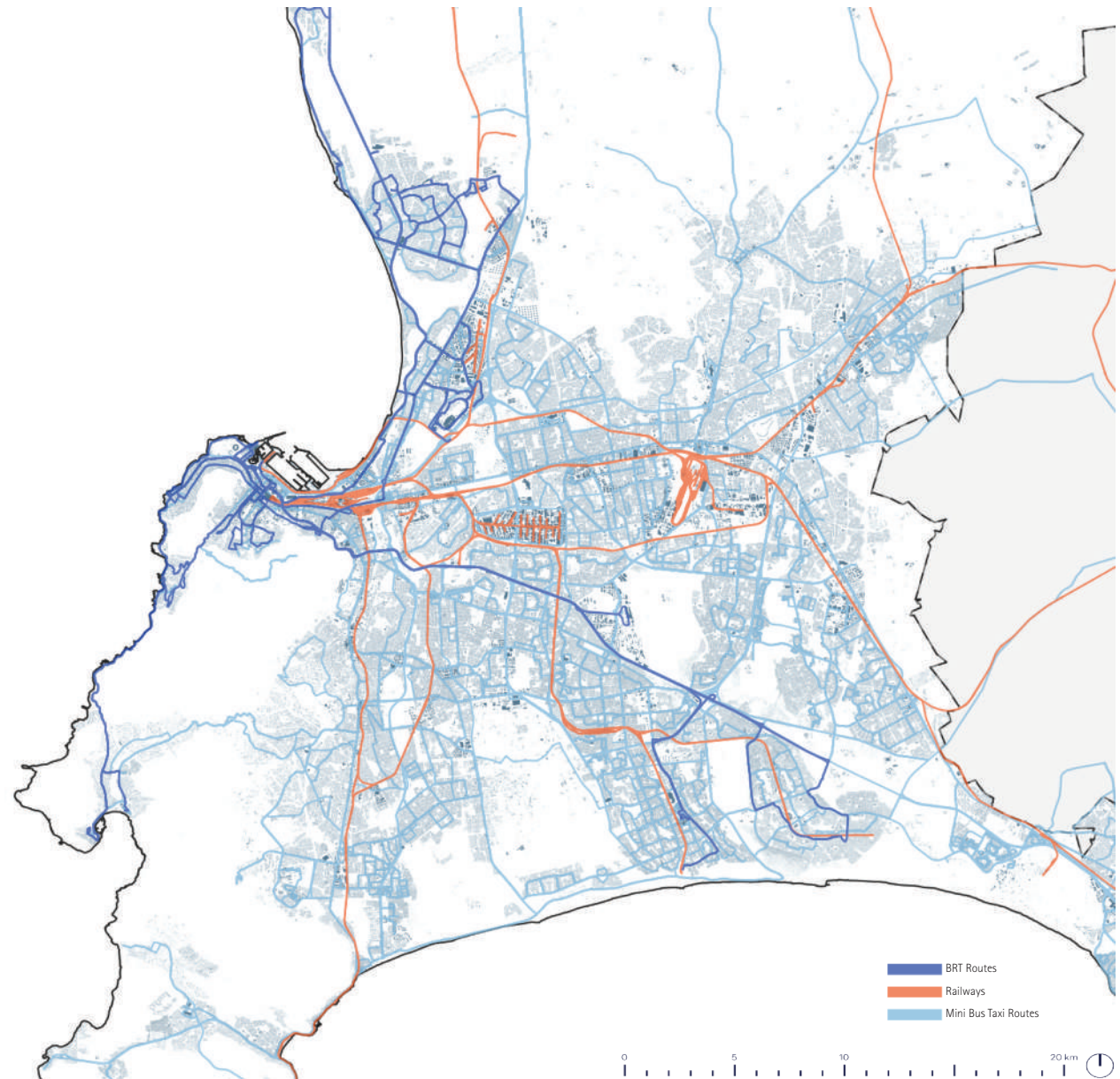
CT ranks among the world's most congested cities, with drivers losing an average of 94 hours per year in traffic.¹⁵ The congestion is particularly severe during peak hours, as employment centers like the Central Business District remain distant from residential areas in the southeast, forcing long commutes along corridors that were never designed for current volumes.

¹² City of Cape Town, 2023

¹³ IOL, September 30, 2024

¹⁴ City of Cape Town, 2023

¹⁵ INRIX, 2024 Global Traffic Scorecard, INRIX Research, 2024



Public Transport System

Sources © [Cape Town] Cohort, Norman Foster Institute. 2025

City of Cape Town. Open Data Portal

2.3 The Economic Imperative for Spatial Integration

While Cape Town demonstrates relative economic strength within the South African context, adding 43,000 new jobs between October and December 2023 with employment increasing by 7.4% year-on-year¹⁶, these gains are not equitably distributed across the urban landscape. Nearly 79% of all net jobs created in South Africa were in the Western Cape, yet township unemployment rates remain dramatically higher than city averages¹⁷). In Khayelitsha, Cape Town's largest township with a population approaching one million, the unemployment rate exceeds 54%, with youth unemployment estimated at over 60%¹⁸. The spatial mismatch between where jobs are created and where unemployed workers reside imposes enormous economic costs not only on individuals and households but also on the city's overall productivity and competitiveness.

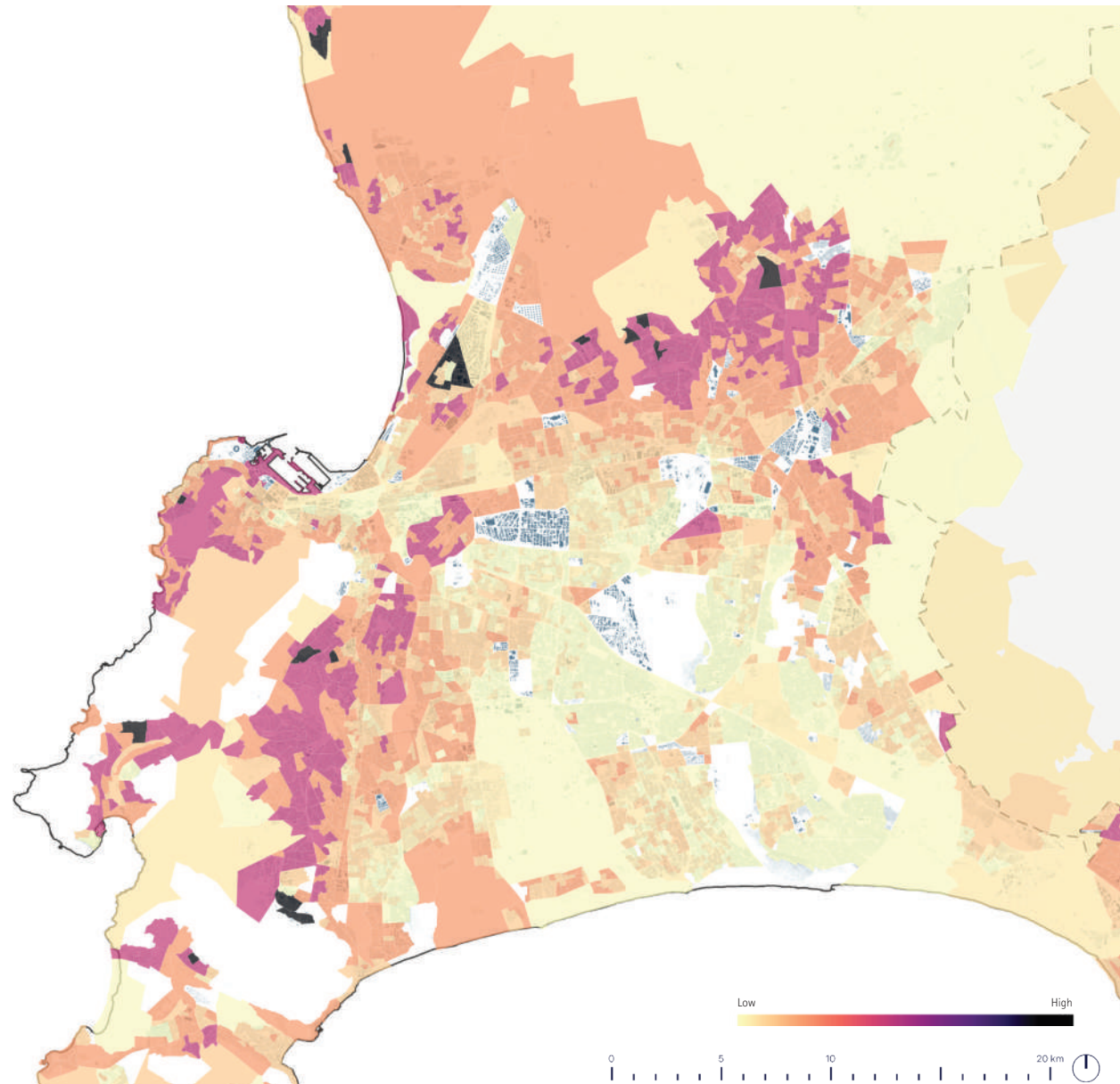
Cape Town's income distribution reveals the depth of economic exclusion. According to the Comprehensive Integrated Transport Plan (CITP), nearly 60% of households earn less than R10,000 per month, with 25% earning R3,500 or less¹⁹. The proportion of households in the lowest income bracket increased from 22.6% in 2019 to 28.0% in 2020, reflecting the pandemic's disproportionate impact on vulnerable populations.²⁰ While the Western Cape maintains South Africa's lowest provincial unemployment rate at approximately 21% in 2025, this figure obscures dramatic variation within the metropolitan area²¹. Township areas experience unemployment rates far exceeding provincial and national averages. Nationally, approximately 40% of the working-age population and 60% of the country's unemployed reside in townships²².

¹⁶ Statistics South Africa, 2024

¹⁷ Western Cape Government, 2024

¹⁸ Baphumelele, n.d.

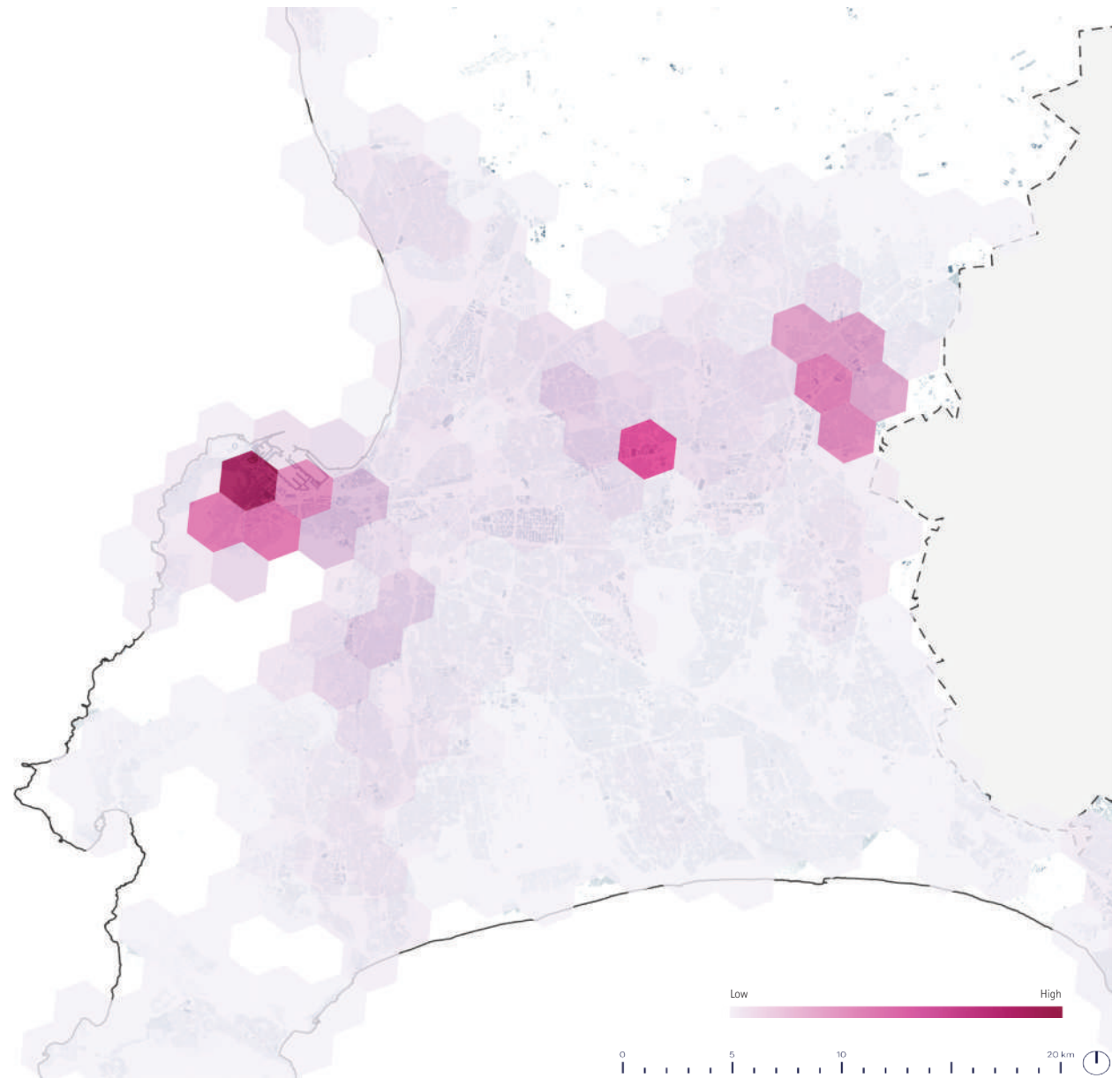
¹⁹ City of Cape Town, 2023



Median Income

Sources © [Cape Town] Cohort, Norman Foster Institute. 2025
City of Cape Town. Open Data Portal

The economic implications of these disparities are substantial. Traffic congestion alone costs the city an estimated R3 billion annually²³, representing not only lost productivity from the 94 hours each driver spends in traffic, but also increased fuel costs, vehicle wear, and broader economic impacts of delayed goods and services. When combined with the opportunity costs of spatial exclusion—where township residents spend a significant portion of their income on transport while being effectively locked out of employment opportunities—the total economic burden of Cape Town's spatial inequality is likely far higher. The Opportunity Hub Network addresses this burden directly by bringing services to where people already congregate rather than requiring additional travel, thereby converting wasted commute and waiting time into productive access to work, learning, services, and markets.



²⁰ City of Cape Town, 2022
²¹ Western Cape Government, 2025
²² UNU-WIDER, 2024
²³ Western Cape Government, 2025

2.4 The Policy Context: From Vision to Implementation

The City of Cape Town's policy frameworks provide strong potential outcomes for the Opportunity Hub Network intervention. The Integrated Development Plan (IDP) 2022–2027 explicitly positions public space, environment, and amenities at the core of the city's transformation agenda²⁴. The plan identifies objectives that the hub network directly addresses: creating pathways to employment; upgrading safety and dignity in public spaces through quality public places standards; reducing commute burdens through proximity-based service delivery; building employment districts in townships through spatial integration initiatives; and enabling incremental densification through support for micro-developers and additional dwelling programmes.

At the national level, the National Development Plan envisions cities where more people live closer to their places of work, with improved public transport and upgraded informal settlements²⁵. The Opportunity Hub network contributes to this vision not by relocating residents but by bringing opportunities to where residents already live and move.

The Comprehensive Integrated Transport Plan (CITP) 2023–2028 introduces the “triple access paradigm,” recognising that improved access can be achieved through mobility (transport), proximity (local opportunities), and digital connectivity (substituting physical trips with online services)²⁶. This multi-dimensional approach provides the theoretical foundation for the hub network's integrated service model.

The CITP also explicitly endorses incremental delivery at scale, favouring many small wins across corridors rather than a few megaprojects, aligning with the Opportunity Hub Network's modular, replicable approach.

Recent judicial interventions further underscore the urgency of spatial transformation. The Constitutional Court's December 2024 judgment in Charnell Commando vs City of Cape Town recognised that the City's failure to provide reasonably located temporary emergency housing perpetuates the legacy of spatial apartheid. This landmark ruling establishes a legal imperative for proximity-based solutions consistent with the Opportunity Hub Network's core principles. The convergence of policy support, judicial mandate, and demonstrated need creates exceptionally favourable conditions for considering this type of intervention.

THE CITY'S 16 OBJECTIVES LINKED TO ITS PRIORITIES AND FOUNDATIONS ARE AS FOLLOWS:	
	ECONOMIC GROWTH OBJ 1: Increased jobs and investment in the Cape Town economy
	BASIC SERVICES OBJ 2: Improved access to quality and reliable basic services OBJ 3: End load-shedding in Cape Town over time OBJ 4: Well-managed and modernised infrastructure to support economic growth
	SAFETY OBJ 5: Effective law enforcement to make communities safer OBJ 6: Strengthen partnerships for safer communities
	HOUSING OBJ 7: Increased supply of affordable, well-located homes OBJ 8: Safer, better-quality homes in informal settlements and backyards over time
	PUBLIC SPACE, ENVIRONMENT AND AMENITIES OBJ 9: Healthy and sustainable environment OBJ 10: Clean and healthy waterways and beaches OBJ 11: Quality and safe parks and recreational facilities supported by community partnerships
	TRANSPORT OBJ 12: A sustainable transport system that is integrated, efficient and provides safe and affordable travel options for all OBJ 13: Safe and quality roads for pedestrians, cyclists and vehicles
	OBJ 14: A resilient city
	OBJ 15: A more spatially integrated and inclusive city
	OBJ 16: A capable and collaborative city government

Integrated Development Plan Objectives
Source © City of Cape Town. Integrated Development Plan 2022–2027.

²⁴ City of Cape Town, 2022

²⁵ National Planning Commission, 2012

²⁶ City of Cape Town, 2023

2.5 The Intervention Logic: Networks, Not Projects

Traditional approaches to addressing spatial inequality in Cape Town have typically focused on singular, large-scale interventions: a new rail line, a housing megaproject, or an industrial park. While these have their place, they suffer from long implementation timeframes, high capital requirements, and limited flexibility to adapt to changing conditions. The Opportunity Hub Network represents a fundamentally different approach: networked, incremental, and adaptive. The network operates on principles derived from successful urban acupuncture initiatives globally, where small-scale, strategic interventions catalyse broader systemic change. By creating a network of interconnected hubs rather than isolated facilities, the intervention generates cumulative effects that exceed the sum of individual components. Each hub serves as both a destination and a node in a larger system, creating redundancy, resilience, and amplified impact.

The network logic distinguishes this intervention from previous attempts at township economic development in several critical ways. First, hubs are positioned at transport nodes where populations already concentrate, applying the last-mile principle that meets people where they naturally flow rather than requiring them to travel to new locations. Second, the intervention integrates functions traditionally delivered through separate programmes: workspace, skills, enterprise support, e-services, and public realm, recognising that spatial disadvantage operates across multiple dimensions simultaneously. Third, the modular construction system enables rapid deployment, prototyping, and learning while maintaining

quality standards. Fourth, community co-design ensures that technical solutions respond to local knowledge and priorities. Fifth, the methodology embraces "intentional obsolescence" as a success metric, designing hubs as catalytic rather than permanent installations whose ultimate success lies in rendering themselves unnecessary as permanent infrastructure emerges.



Bus terminal at Station Deck in the Central Business District

Source © [Cape Town] Cohort, Norman Foster Institute. 2025

2.6 The Research Commission

This research responds to two complementary commissions that together frame both the analytical challenge and implementation pathway for Cape Town's spatial transformation. The City of Cape Town's formal research brief, issued through the masters programme "Sustainable Cities", provides the metropolitan-scale framework. The relationship with the Mission for Inner City provides insights into Cape Town's implementation mechanisms. Together, they create a unique opportunity to bridge the persistent gap between urban analysis and on-ground action.

The City's formal commission, titled "Provision of Recreational Open Space and Public Space in Cape Town," directly supports the review of the Metropolitan Spatial Development Framework (MSDF) and development of a comprehensive public space project pipeline for the Quality Public Places Programme (QPPP). This brief poses fundamental questions about spatial equity and dignity:

Is Cape Town's provision of recreational open space sufficient to address future needs and ensure a dignified, sustainable, and healthy quality of life for its residents?

How does Cape Town's provision compare to other cities facing similar challenges of rapid urbanisation with resource constraints?

What can Cape Town learn from leading sustainable cities when planning green infrastructure?

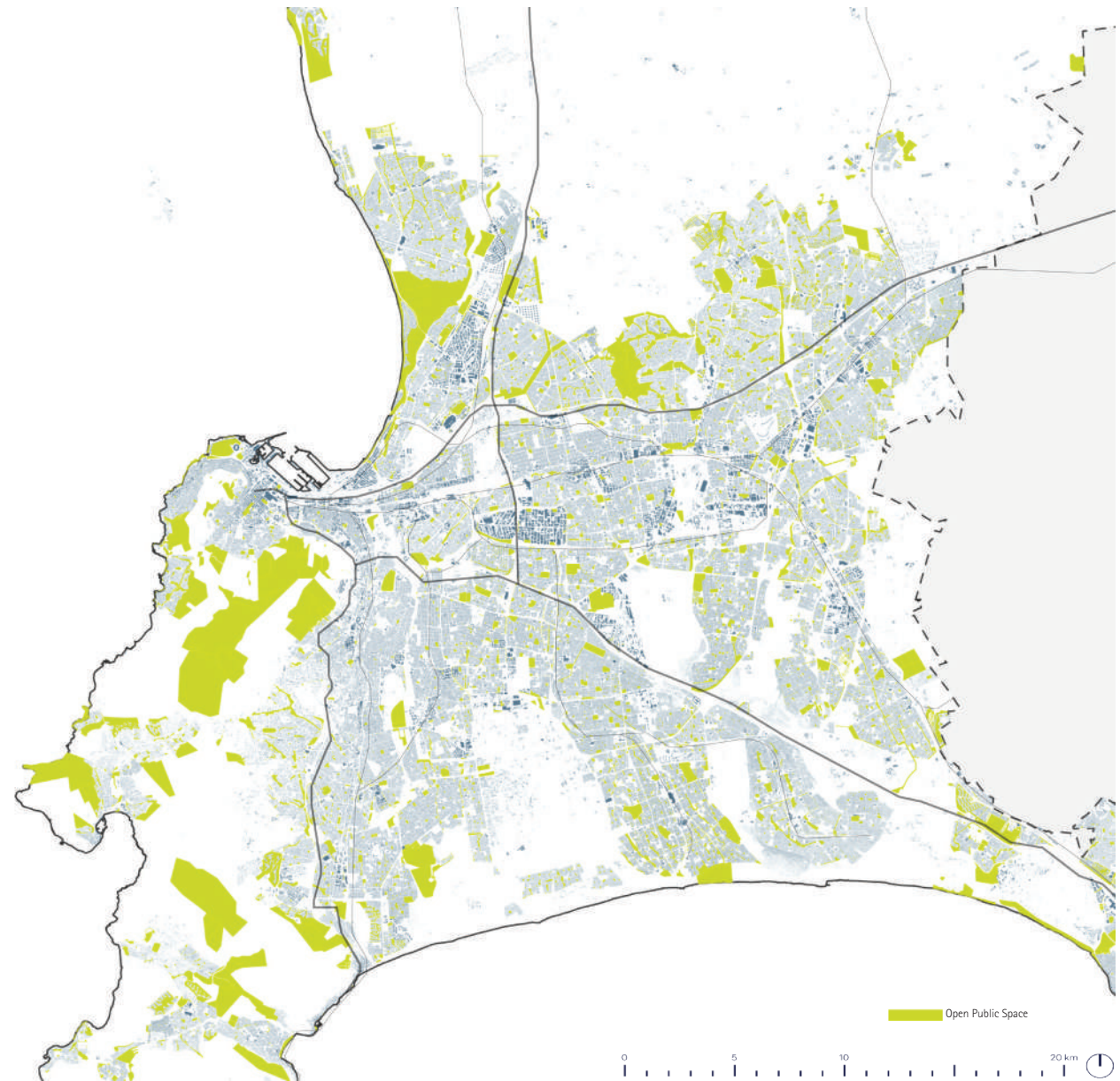
The brief explicitly identifies the Cape Town CBD as a key project area and requires assessment at metropolitan, district, and local levels.

Complementing the City's analytical requirements, the Mission for Inner City Cape Town provides the implementation vehicle for translating research into action. Their placemaking vision, to make Cape Town's Inner City "a place where anyone can walk safely, easily, and enjoyably any time of day or night," operationalises the QPPP's broader objectives through four critical themes: safety, addressing the temporal dimension of public space where relative daytime safety gives way to nighttime hostility; walkability, challenging car-centric infrastructure despite high pedestrian traffic; comfort, responding to Cape Town's extreme seasonal variations and climate vulnerability; and conviviality, transforming historically transient and transactional spaces into places that invite genuine public life.

The intervention builds on research conducted through the programme, including spatial analysis of public space provision, mapping of transport accessibility, and community engagement exercises. The Violence Prevention through Urban Upgrading (VPUU) programme's 15-year experience in Cape Town provides crucial evidence for the effectiveness of integrated spatial interventions, demonstrating a 40% reduction in interpersonal violence through strategic public space upgrades²⁷. This evidence base informs the Opportunity Hub Network's emphasis on public realm quality alongside service provision, recognising that dignified public spaces generate benefits that extend far beyond their immediate amenity value.

Global evidence from similar networked intervention models demonstrates that strategic, networked interventions can achieve transformative impacts when aligned adequately with local conditions and policy frameworks. Medellín's Unidades de Vida Articulada (UVAs), which transformed water reservoirs into community spaces serving underserved neighbourhoods, illustrate how modest infrastructure investments can catalyse broader social transformation. Barcelona's network of community facilities demonstrates scaling principles for distributed service delivery. Kigali's innovation hubs show how workspace and enterprise support can be delivered effectively in resource-constrained contexts. These international precedents inform the Opportunity Hub Network's design while recognising that successful implementation requires adaptation to Cape Town's specific conditions of transport-dependent townships and existing municipal partnerships.

²⁷ VPUU, 2024



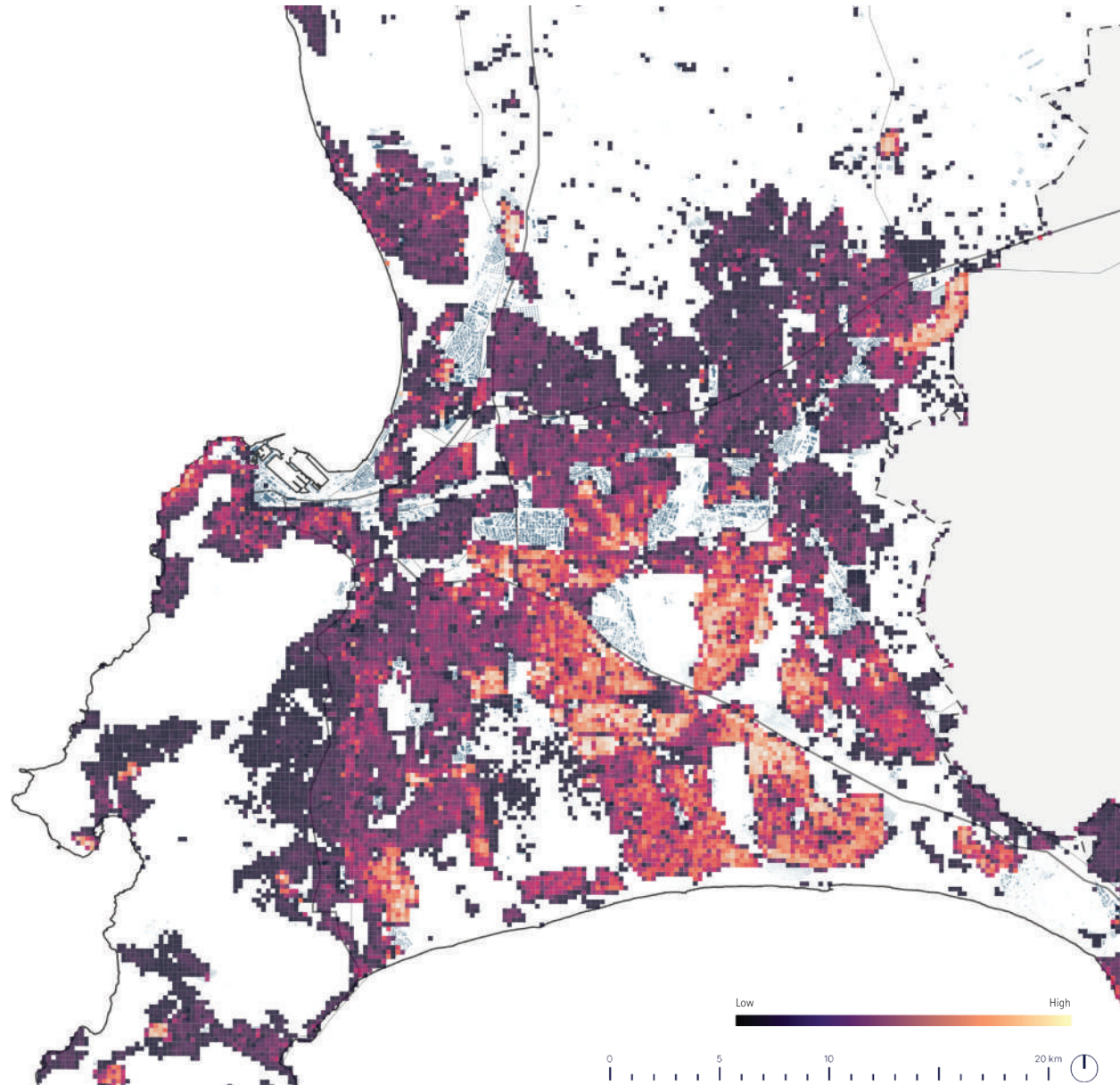
Classified Open Public Space

Sources © [Cape Town] Cohort, Norman Foster Institute. 2025

City of Cape Town. Open Data Portal

2.3 The Moment for Bold Action

Cape Town finds itself at a pivotal moment. The city is experiencing rapid demographic change, primarily driven by job-seeking migrants from the Eastern Cape settling in townships like Khayelitsha, Gugulethu, and Nyanga, with these historically underserved areas now demanding substantial infrastructure investment²⁸. The city can either continue with incremental, disconnected interventions that fail to address systemic spatial inequality, or it can embrace bold, networked solutions that transform the geography of opportunity.



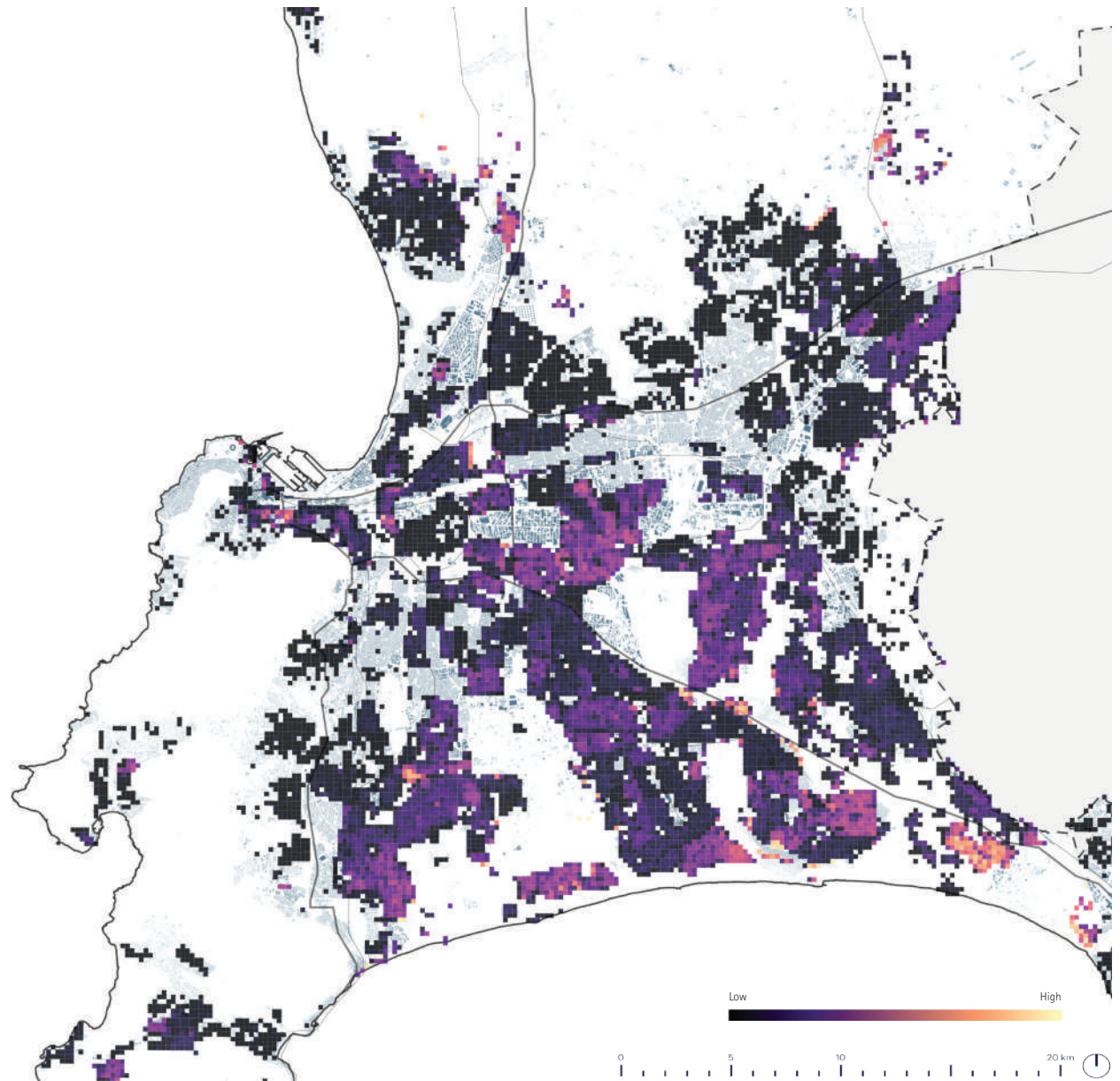
²⁸ Newsday, 2025

Population Density 2020

Sources © [Cape Town] Cohort, Norman Foster Institute. 2025

City of Cape Town. Open Data Portal

The Opportunity Hub Network represents such a bold solution, one that is simultaneously ambitious in vision and pragmatic in implementation. By starting with a focused pilot and considering a systematic scale-up to 120 hubs by 2050, this approach suggests a potential pathway from Cape Town's divided present toward a more integrated future, where opportunity could become more accessible regardless of residential location. The evidence is clear, the policy framework is aligned, and the implementation pathway is defined. What remains is the political will and institutional commitment to transform Cape Town from a city where geography determines destiny to one where opportunity is truly democratised.



Population Increase Projection 2040

Sources © [Cape Town] Cohort, Norman Foster Institute. 2025

City of Cape Town. Open Data Portal

3. Problem Analysis

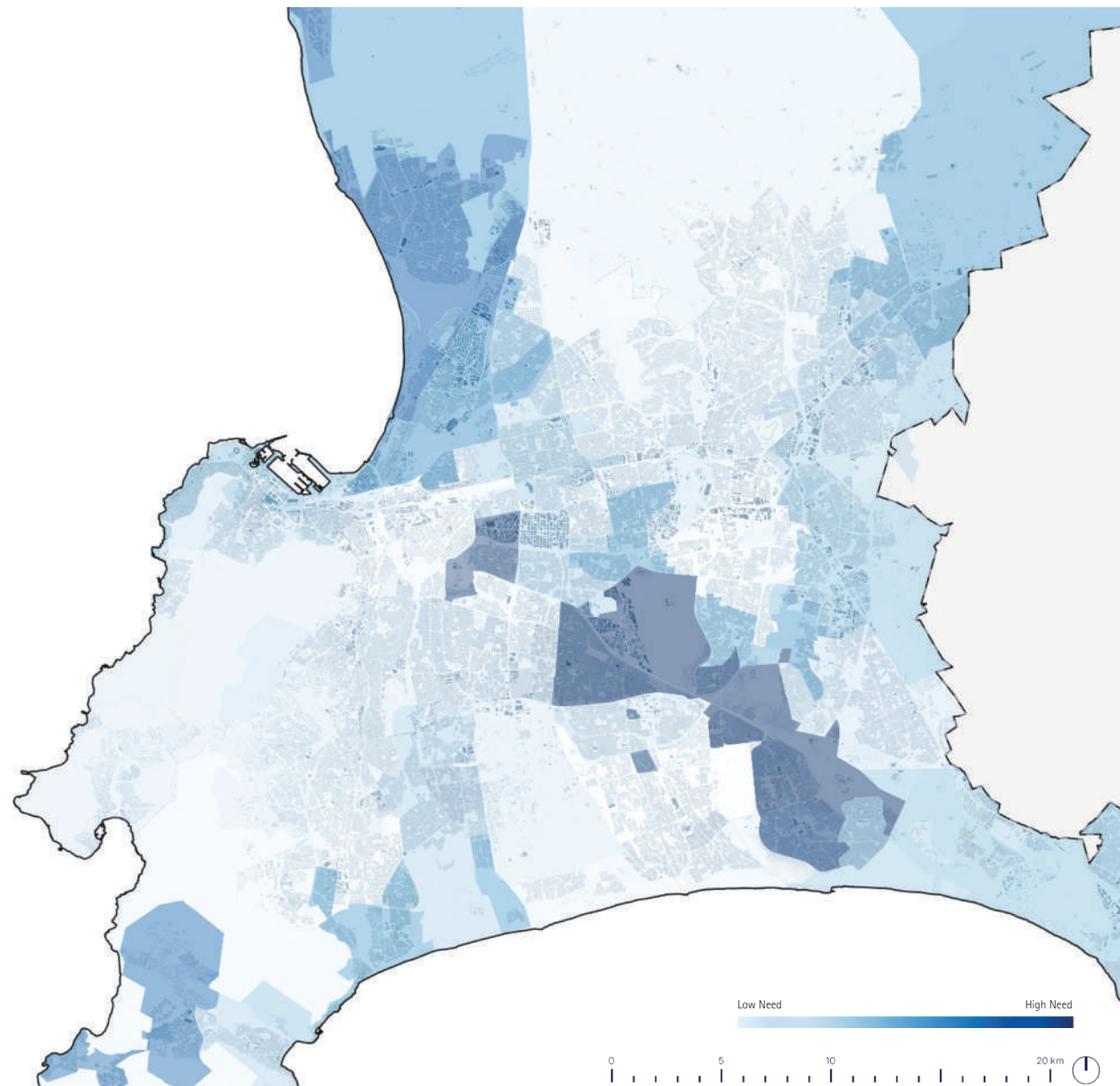
3.1 Housing and Basic Services

Cape Town's housing crisis reflects the same spatial logic. Approximately 146,000 households reside in 437 informal settlement pockets across the city, many established before democracy but still lacking formal recognition and security of tenure²⁹. An additional significant population lives in "backyard dwellings," informal structures on the same property as formal homes, without direct access to services. The 2024 Cape Town Household Survey reveals that substantial proportions of households in informal dwellings lack adequate access to water, sanitation, and electricity.

The waiting list for council housing exceeds 350,000 people in Cape Town alone, with some applicants having waited since before the democratic transition³⁰. The pace of formal housing delivery has not kept up with population growth and urbanization, leading to continued expansion of informal settlements despite significant state investment.

Basic services deficits concentrate in the same areas experiencing transport burdens and economic exclusion. Households in informal settlements and backyard dwellings often rely on communal water points, bucket toilets or shared sanitation facilities, and unsafe energy sources for cooking and heating. These conditions impose health risks and dignity costs that compound other dimensions of disadvantage.

²⁹ ISMaps, n.d.
³⁰ Al Jazeera, 2024



Basic Services, showing no piped water (2022)

Sources © [Cape Town] Cohort, Norman Foster Institute. 2025
City of Cape Town. Open Data Portal

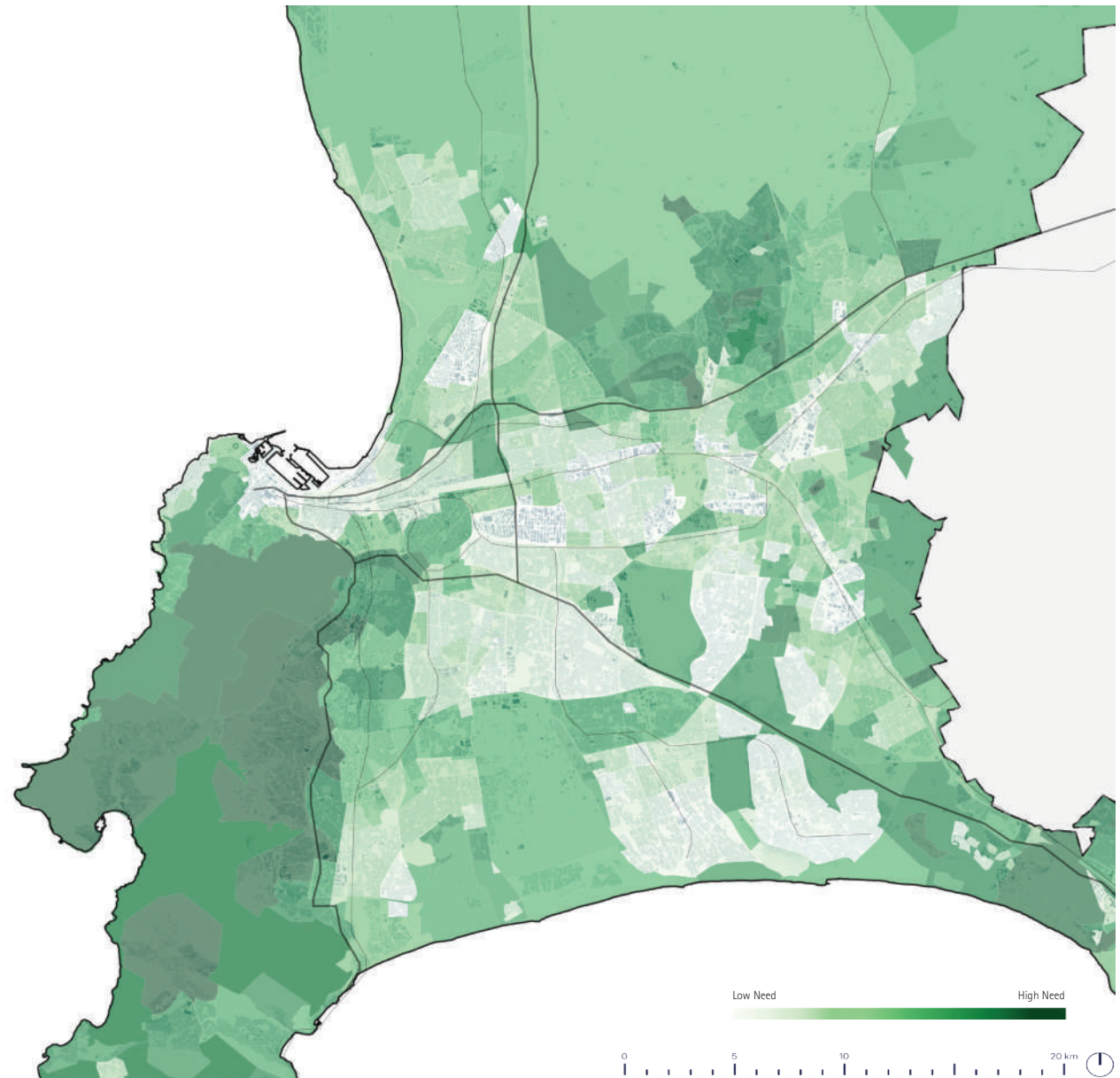
3.2 Access to Public Space and Services

The distribution of public space reflects and reinforces existing inequalities. Research indicates that only 25% of lower-income households live within walking distance of quality public spaces³¹. Well-maintained parks, plazas, and recreational facilities cluster in historically white suburbs, while township public spaces suffer from inadequate maintenance, poor lighting, and safety concerns.

This disparity extends to civic services. Libraries, municipal offices, police stations, and healthcare facilities concentrate in areas accessible to private vehicle users but distant from public transport-dependent populations. Residents seeking to complete administrative tasks, renew documentation, or access government services often face full-day journeys that consume wages and require taking time from work.

The City of Cape Town's event economy illustrates the potential value of activated public space: Cape Town recorded nearly R4 billion from major events in 2023, creating 26,000 jobs. Yet these benefits concentrate in well-resourced venues while township public spaces, despite hosting vibrant informal economic activity, lack the infrastructure investment to realize similar value.

³¹ Stockholm Resilience Centre, 2020



Green Ratio (2024)

Sources © [Cape Town] Cohort, Norman Foster Institute. 2025

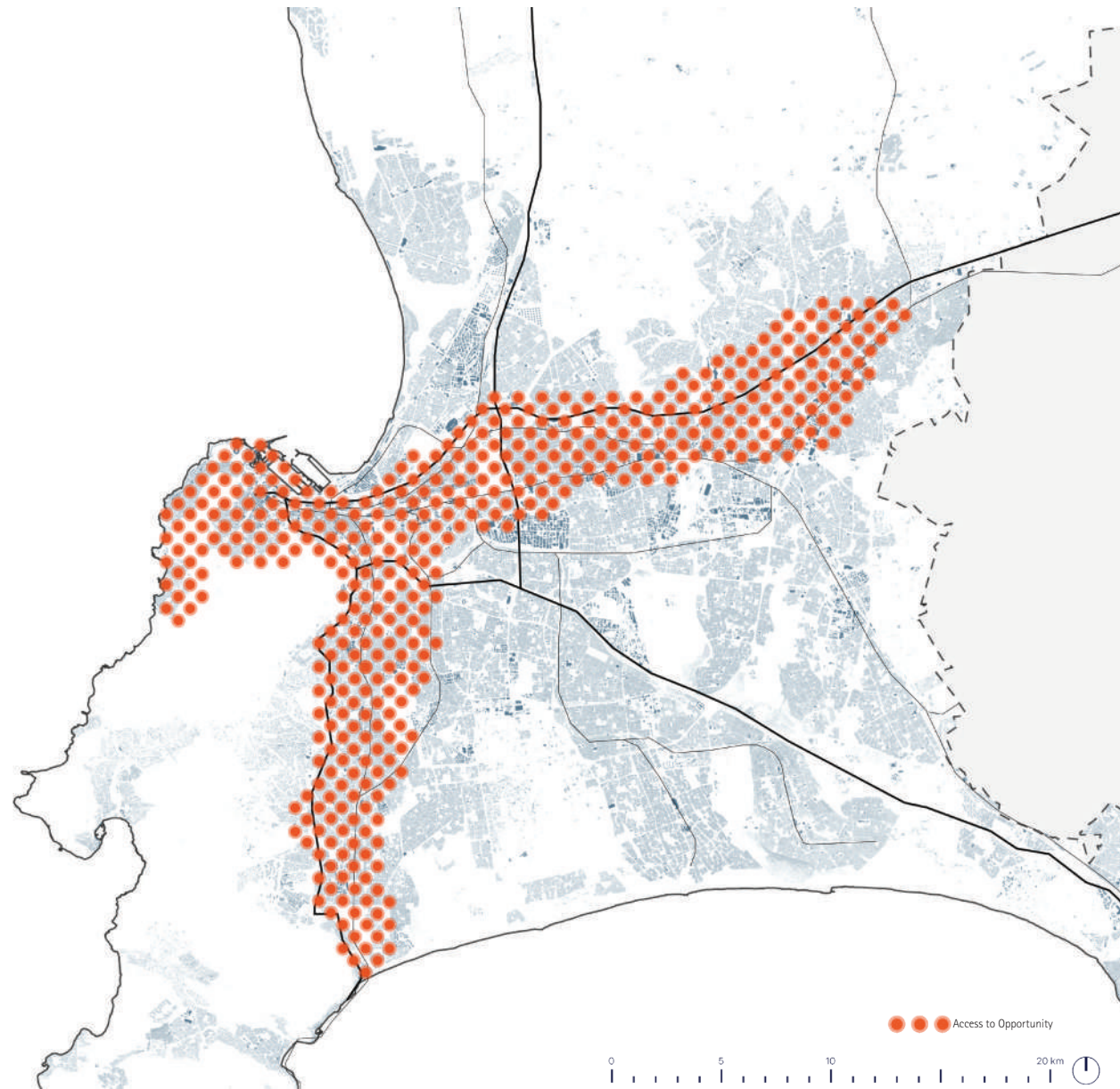
City of Cape Town. Open Data Portal

3.3 The Compounding Nature of Spatial Disadvantage

These individual deficits do not operate in isolation; they interact to create reinforcing cycles of exclusion. A resident of Khayelitsha seeking employment in the CBD faces a 35-kilometre journey consuming two hours and a significant portion of any potential wages. The same resident lacks access to digital connectivity for job searching, faces barriers to obtaining required documentation due to distant civic services, and returns home to informal housing lacking the electricity needed for evening study or rest adequate for the following day's commute.

This compounding dynamic explains why spatial inequality has proven so persistent despite significant post-apartheid investment. Interventions addressing single dimensions, whether housing or transport or services, cannot overcome disadvantages that operate across all dimensions simultaneously. The problem analysis therefore points toward a different approach: place-based interventions that address multiple dimensions of opportunity deficit at locations where they can reach the populations most affected.

The transport nodes where township residents wait for Mini Bus Taxis, Trains or Buses, transfer between modes, and begin their daily journeys into the formal economy represent precisely such locations. These are places where significant populations already congregate, where time currently lost to waiting could be transformed into productive activity, and where services could intercept daily flows rather than requiring additional travel.



Opportunity Corridor

Sources © [Cape Town] Cohort, Norman Foster Institute. 2025

City of Cape Town. Open Data Portal



St. Georges Mall, Central Business District and a township in Cape Town

Sources © Left: Oberholzer, Bernard Et van Papendorp, Johan.

Right: Malume, Rich.

4. Conceptual Framework



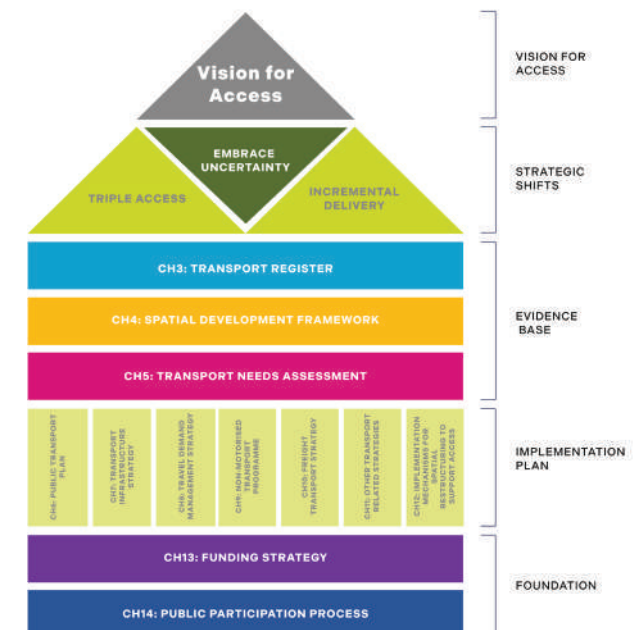
Bree Street Sundays
Source © Cuniffe, Teagan.

4.1 Defining Opportunity in the Cape Town Context

The Opportunity Hub intervention rests on a particular understanding of what "opportunity" means for residents of Cape Town's townships. Rather than measuring well-being solely through income or resource provision, this research adopts a capabilities-based framework that asks what people are actually able to do and become. This approach, developed by economist and philosopher Amartya Sen, argues that development should be understood as the expansion of human capabilities rather than simply economic growth³². Poverty, in this view, is not merely the absence of income but the deprivation of capabilities necessary to live a life one has reason to value.

The capability approach proves particularly relevant to Cape Town's spatial challenges. As the Oxford Poverty and Human Development Initiative notes, Sen's framework recognizes that the same resources can yield vastly different outcomes depending on circumstances: the same quantity of food may meet the needs of a child but not an agricultural labourer; the same transport fare represents different burdens for households at different income levels³³. In a city where apartheid planning deliberately concentrated disadvantage in peripheral locations, the capability to access services, employment, and civic participation depends not only on what resources exist but on where they are located relative to where people live.

This conceptualization aligns with the City of Cape Town's own framing. The Comprehensive Integrated Transport Plan (CITP) 2023–2028 articulates a vision in which "all people have efficient access to a range of opportunities in a manner that is sustainable and provides dignity"³⁴. The CITP explicitly connects transport to capabilities: access to work and education, to information and services, to goods and social connection. The Opportunity Hub intervention operationalizes this vision by identifying where capability deprivations concentrate and designing interventions that expand real freedoms at those locations.



CITP Objectives

Source © City of Cape Town. Comprehensive Integrated Transport Plan 2023–2028.

³² Sen, 1999

³³ OPHI, n.d.

³⁴ City of Cape Town, 2023, p. 11

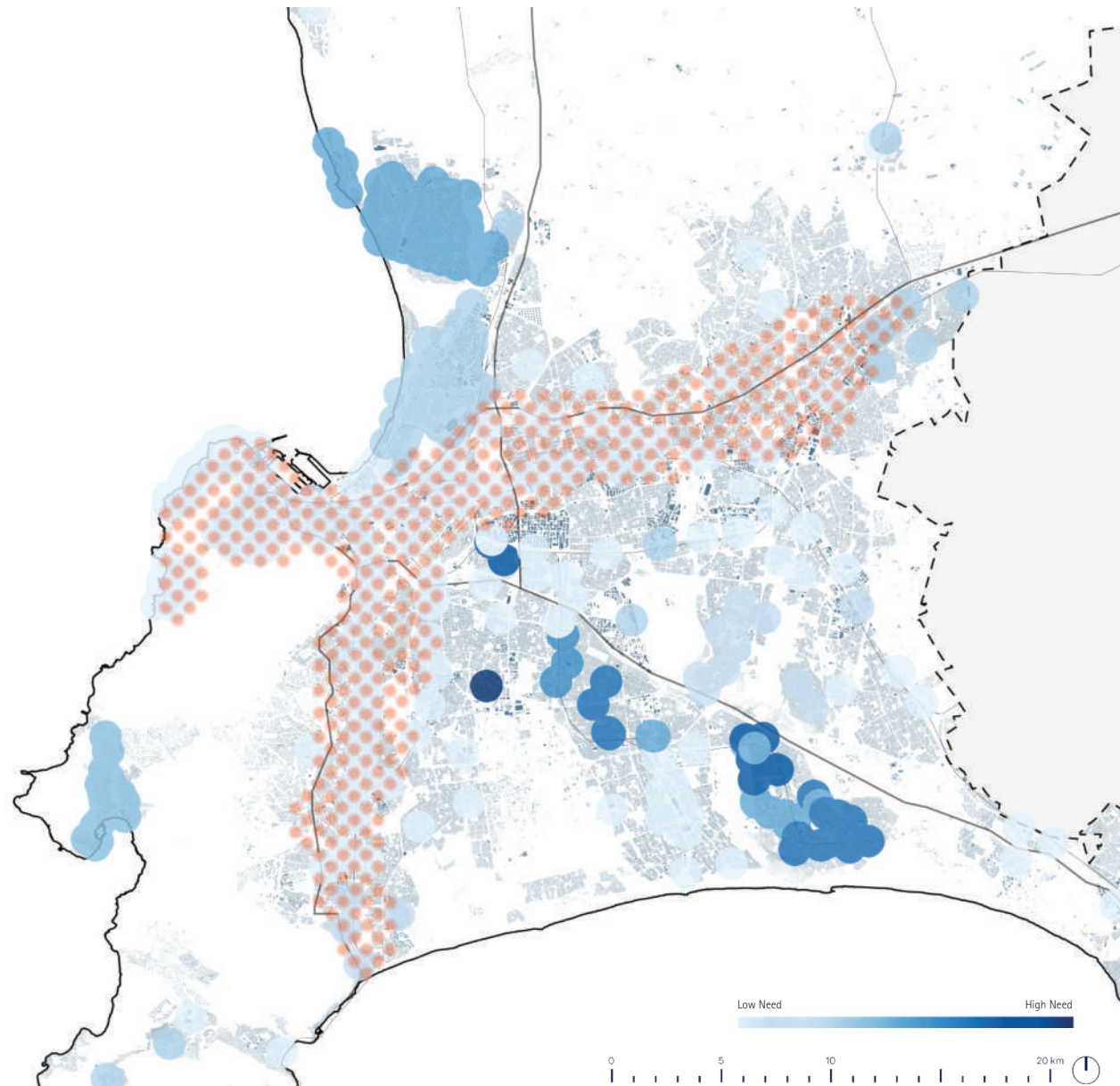
4.2 Five Dimensions

Translating the capabilities framework into measurable indicators requires specifying which capabilities matter most in the Cape Town context. The research identified five dimensions through analysis of local policy priorities, national development frameworks, and the specific constraints facing township residents:

Basic Services

addresses what Sen termed "elementary capabilities" necessary for survival and dignified living. The South African Constitution enshrines rights to water, sanitation, and energy; the National Development Plan commits to upgrading all informal settlements by 2030³⁵. Yet significant portions of the population lack reliable access to these fundamentals. This dimension measures the gap between constitutional rights and lived reality.

³⁵ National Planning Commission, 2012

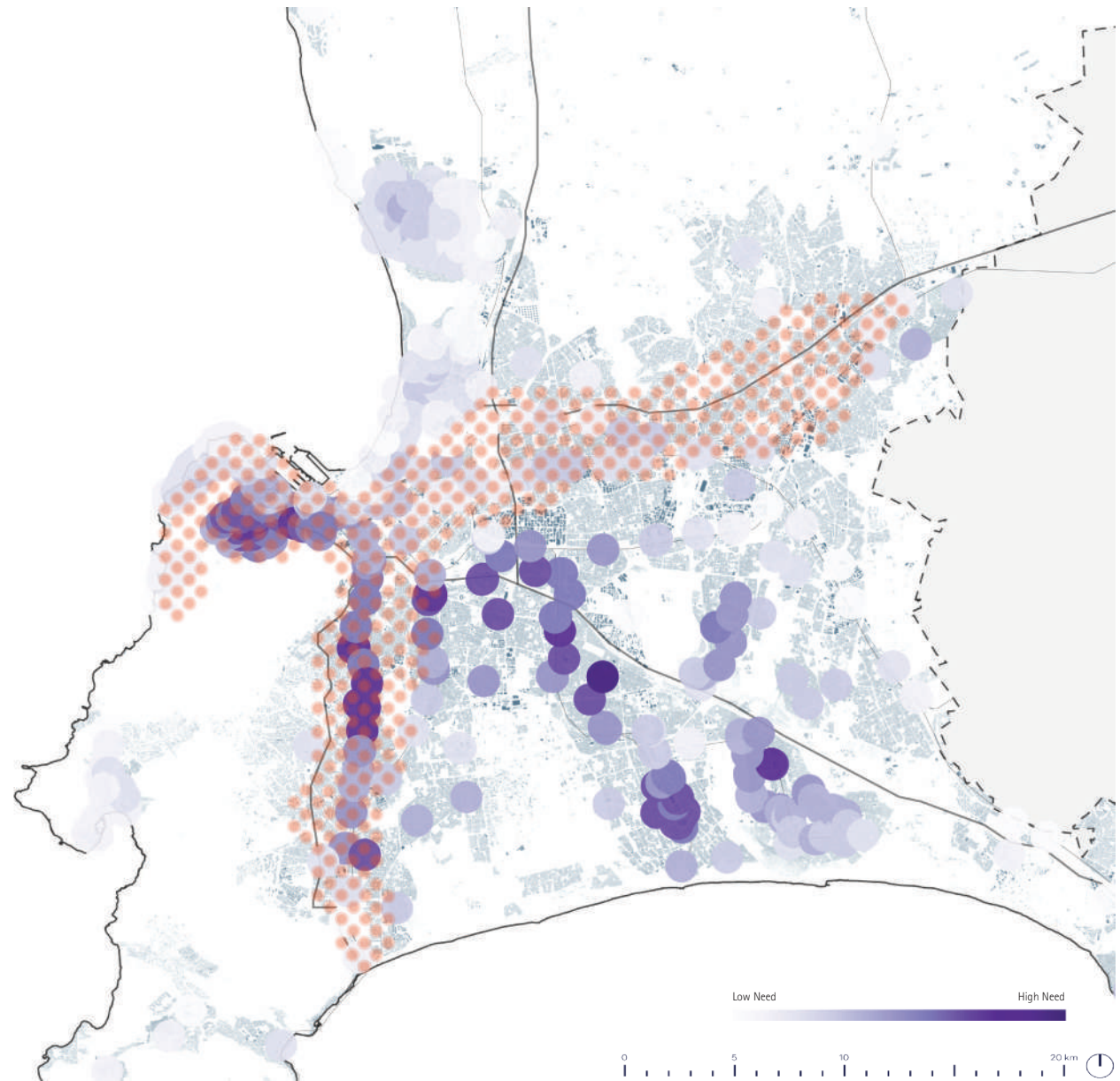


Transport Nodes showing aggregated Data: no piped water (2022)

Sources © [Cape Town] Cohort, Norman Foster Institute. 2025
City of Cape Town. Open Data Portal

Learning and Skills

responds to the NDP's emphasis on education as the primary mechanism for breaking intergenerational poverty. The plan identifies human capabilities, particularly education and skills development, as essential for both individual advancement and national economic transformation³⁶. Youth unemployment exceeding 60% in many townships indicates that learning opportunities must extend beyond formal schooling to include practical skills aligned with labour market demand.



³⁶ National Planning Commission, 2012

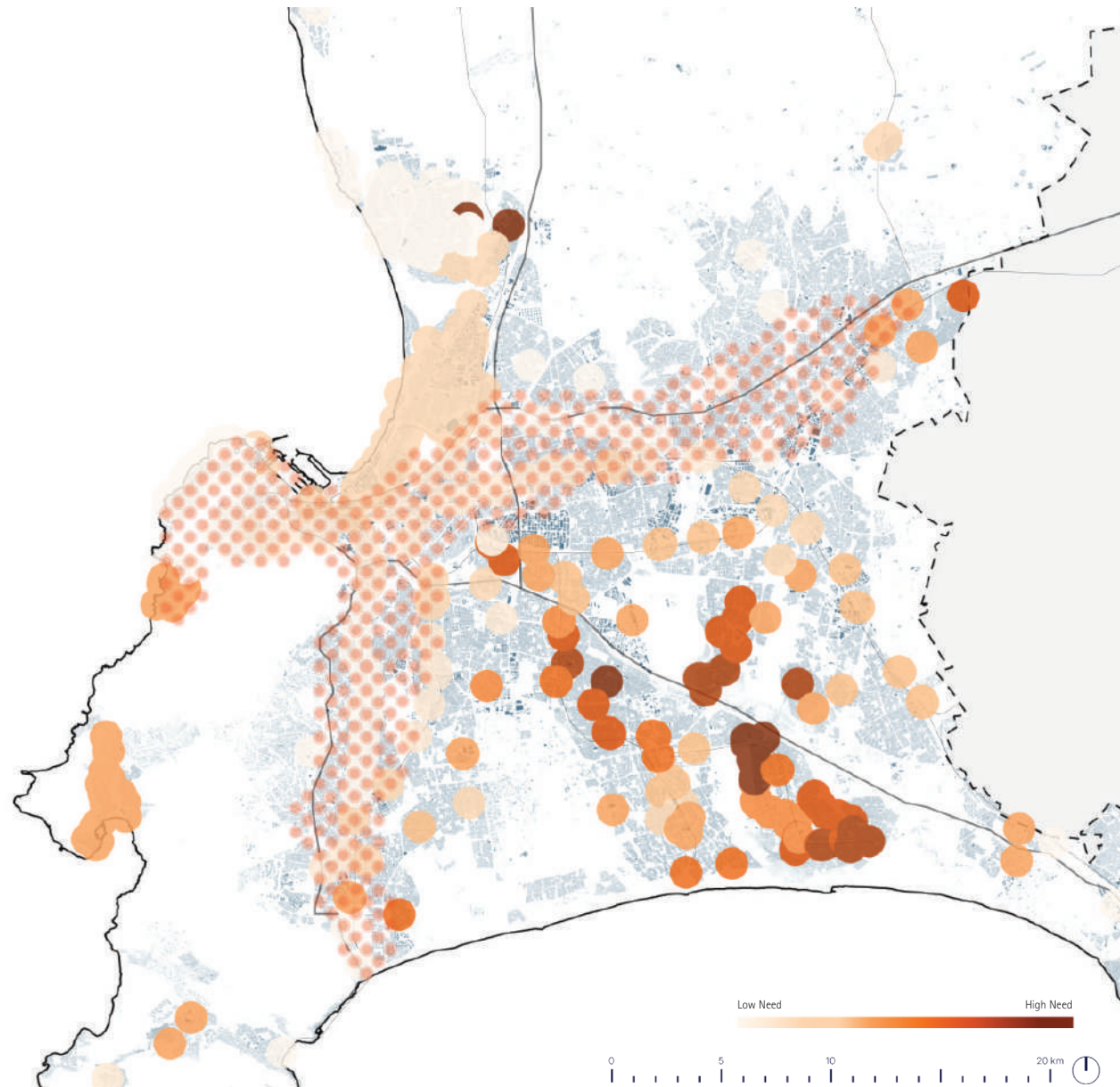
Transport Nodes showing aggregated Data: Access to Schools (2022)

Sources © [Cape Town] Cohort, Norman Foster Institute. 2025

City of Cape Town. Open Data Portal

Work and Enterprise

recognizes that spatial inequality in Cape Town manifests acutely as separation between where people live and where jobs exist. The City's Integrated Development Plan commits to "plan for employment and improve accessibility and access to economic opportunities," acknowledging that spatial form directly affects economic inclusion³⁷. This dimension captures unemployment, income levels, and dependency ratios that indicate economic exclusion.

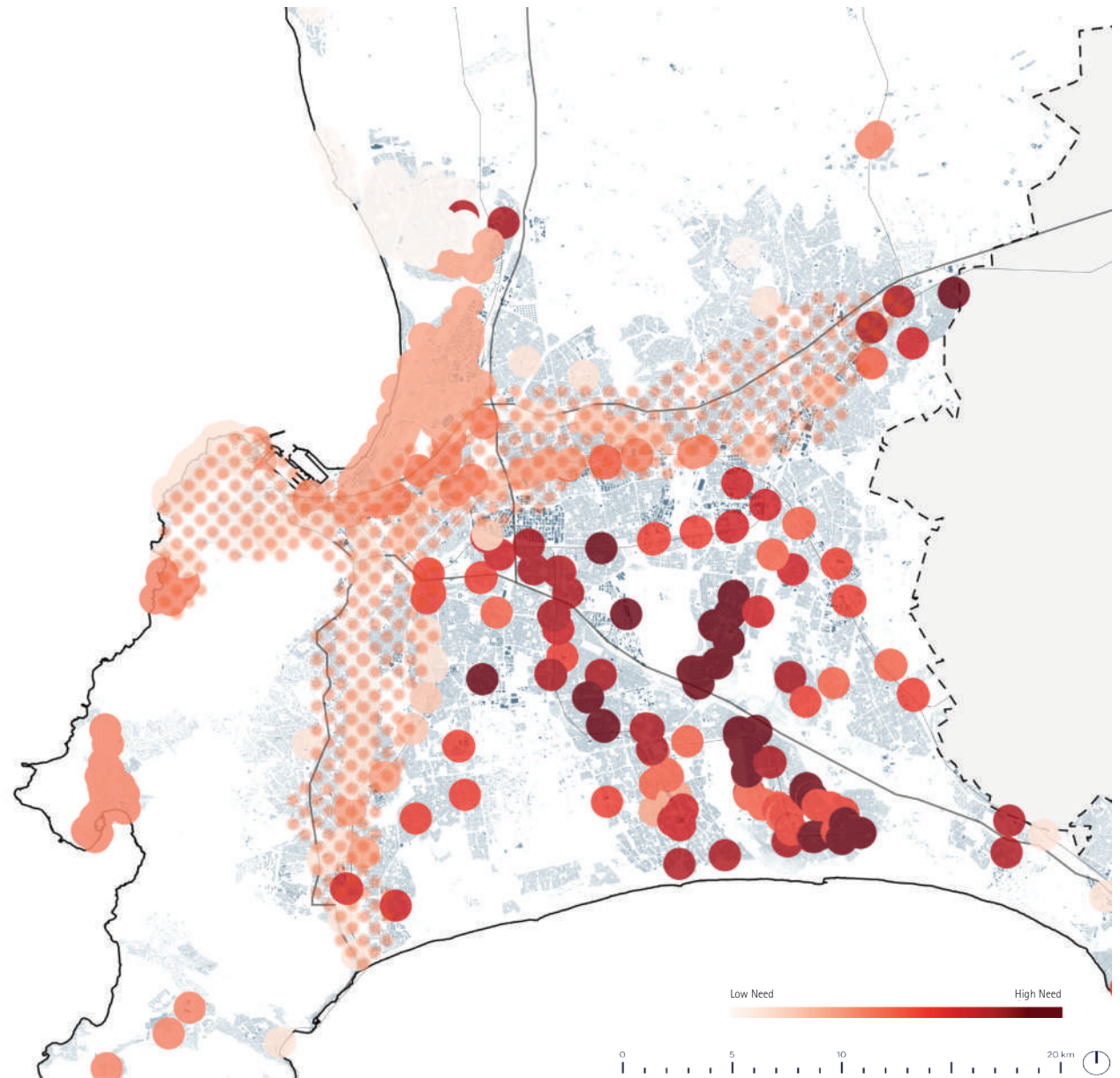


³⁷ City of Cape Town, 2022, p. 43

Transport Nodes showing aggregated Data: Unemployment (2022)
Sources © [Cape Town] Cohort, Norman Foster Institute. 2025
City of Cape Town. Open Data Portal

Digital and Media

reflects the CITP's "triple access paradigm," which identifies digital connectivity alongside physical mobility and spatial proximity as pathways to opportunity³⁸. In an economy increasingly structured around information technology, households without internet, computers, or mobile phones face compounding exclusion from services, employment, and civic participation.



³⁸ City of Cape Town, 2023

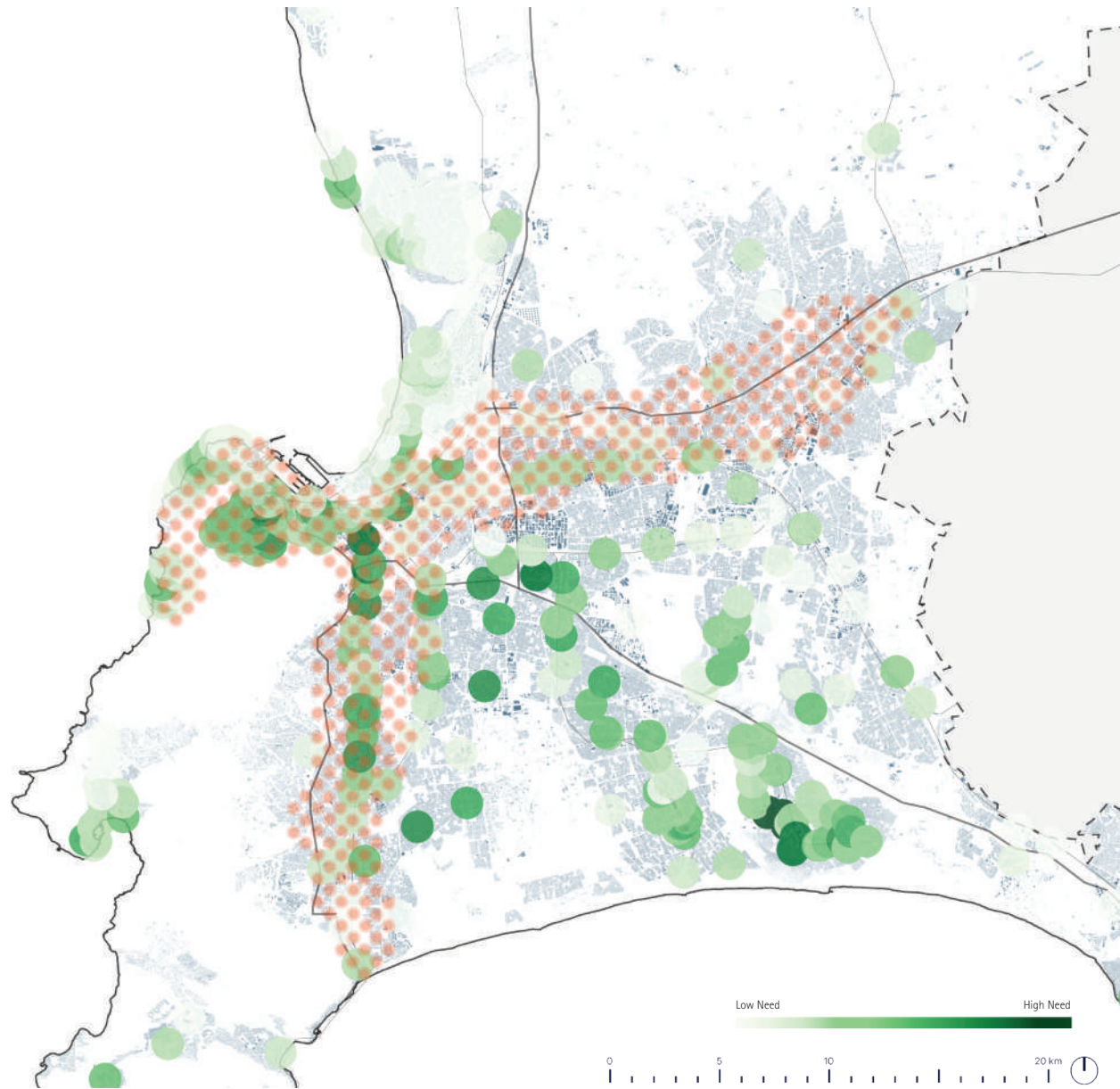
Transport Nodes showing aggregated Data: No access to Wifi (2022)

Sources © [Cape Town] Cohort, Norman Foster Institute. 2025

City of Cape Town. Open Data Portal

Civic and Administration

addresses what might be termed "administrative capabilities": the ability to access government services, obtain documentation, and participate in civic life. Administrative exclusion compounds other forms of disadvantage; residents unable to reach municipal offices, libraries, or police stations face barriers to licensing, reporting, and the formal procedures that economic participation requires.



Transport Nodes showing aggregated Data: Access to Libraries (2022)

Sources © [Cape Town] Cohort, Norman Foster Institute. 2025

City of Cape Town. Open Data Portal

These five dimensions collectively capture what the capability approach terms "functionings": the beings and doings that constitute a person's life³⁹. The framework does not prescribe which combination of functions people should pursue but rather measures whether residents have real opportunities to pursue the lives they value.

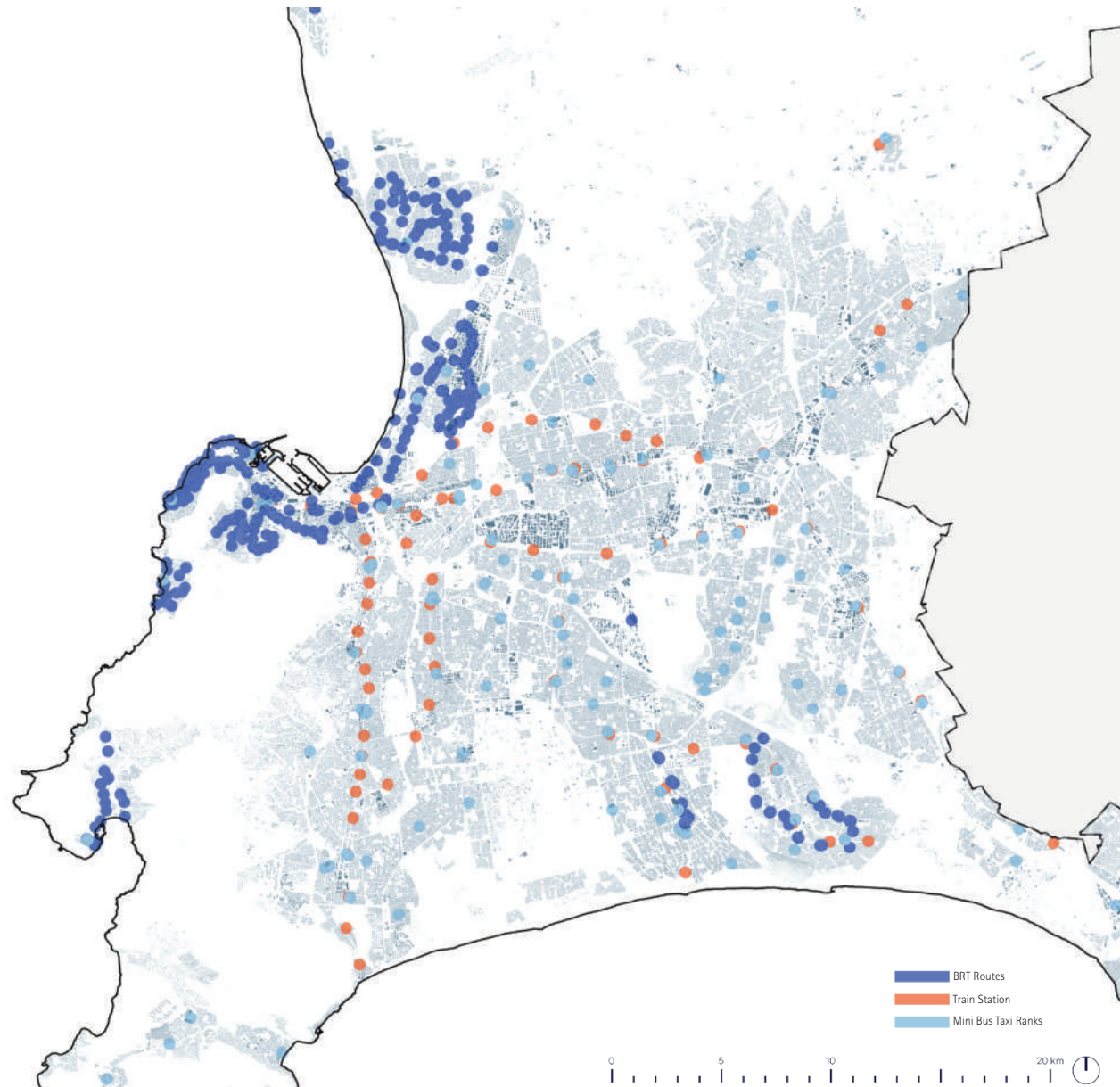
³⁹ Robeyns, 2016

4.3 Transport Nodes as Capability Multipliers

The decision to locate interventions at transport nodes reflects both theoretical reasoning and practical observation. Transit-oriented development research demonstrates that well-designed interventions at transport nodes can improve access to jobs and services while enhancing neighbourhood livability⁴⁰. Emerging literature on equitable TOD specifically examines how transit investments can either exacerbate or reduce inequality depending on implementation approach⁴¹.

For Cape Town's township residents, transport nodes function as what the research team terms "capability multipliers": locations where improvements yield benefits across multiple dimensions simultaneously. A commuter accessing services at a transport hub during waiting time gains capability without additional travel cost or time expenditure. This insight, formalized as the "last mile principle," inverts conventional service delivery logic by embedding services within existing movement patterns rather than expecting residents to travel to them.

The CITP validates this approach, noting that Cape Town should "increase access via mobility (PT/NMT), proximity (local opportunities), and digital (e-services, online work)" and that interventions combining all three in one location represent optimal policy⁴². Transport nodes naturally satisfy these criteria, making them strategic sites for multi-functional services.



⁴⁰ Cervero, 2004

⁴¹ Debrunner & Gerber, 2024

⁴² City of Cape Town, 2023

Transport Nodes, categorized

Sources © [Cape Town] Cohort, Norman Foster Institute. 2025

City of Cape Town. Open Data Portal

OpenStreetMap.

4.4 Guiding Principles

Three principles structure how the Opportunity Hub concept translates from theory to implementation:

Accessibility

requires that interventions reach people experiencing capability deprivation without imposing additional burdens. The catchment area adopted in site selection reflects pedestrian accessibility standards while acknowledging that township residents should not bear further transport costs to access basic services. This principle also encompasses universal design considerations ensuring that hubs serve residents regardless of physical ability.

Modularity

enables the balance between standardization and customization that large-scale implementation requires. Research on modular construction systems demonstrates that standardized components can effectively enable site-specific adaptation⁴³. The Opportunity Hub catalogue matrix operationalizes this principle: structural frames and common programmes provide consistency, while specific programming and finishing respond to local conditions and community preferences.

Community Integration

ensures that interventions strengthen rather than disrupt existing social infrastructure. The participation methodology treats communities as co-designers rather than consultants, recognizing that residents possess knowledge of local dynamics that quantitative analysis cannot capture. This principle also requires that hubs integrate with informal economic activity, existing gathering patterns, and neighbourhood identity rather than imposing external spatial logics.

⁴³ Lidelöw, 2024

5. Research Findings and Analytical Framework

5.1 Rationale for a Data-Driven Approach

Addressing Cape Town's spatial inequality requires moving beyond intuition to establish an evidence base that guides investment decisions across hundreds of potential intervention sites. The research underpinning the Opportunity Hub intervention began with a fundamental question: how can opportunity deficits be systematically measured, compared, and prioritized across a metropolitan region of 4.6 million people? The answer required developing an analytical framework that translates abstract concepts of inequality into quantifiable metrics, then applying this framework consistently across the city's transport network.

The findings presented here explain why transport nodes emerged as the optimal intervention typology, how the five Key Performance Indicators were derived, and what the citywide analysis revealed about the geography of opportunity in Cape Town. These insights directly shaped the design parameters, programmatic priorities, and site selection criteria that structure the Opportunity Hub approach.

KPIs

Basic Services

1. Informality
2. Toilet Insecurity
3. Light Energy Insecurity
4. Heat Energy Insecurity
5. Piped Water Insecurity
6. Hospitals within 5km

Learning and Skills

1. Low Higher education
2. Schools in 5km
3. Youth Population Density

Work and Enterprise

1. Unemployment ratio
2. Low Median Income Level
3. Household Dependency Ratio

Digital and Media

1. Lack of Internet access
2. No cellphone access
3. No Computer access

Civic and Admin

1. Libraries in 5km
2. Police stations in 5km
3. City offices in 5km
4. Post Office 5km

Transport Nodes in Capetown



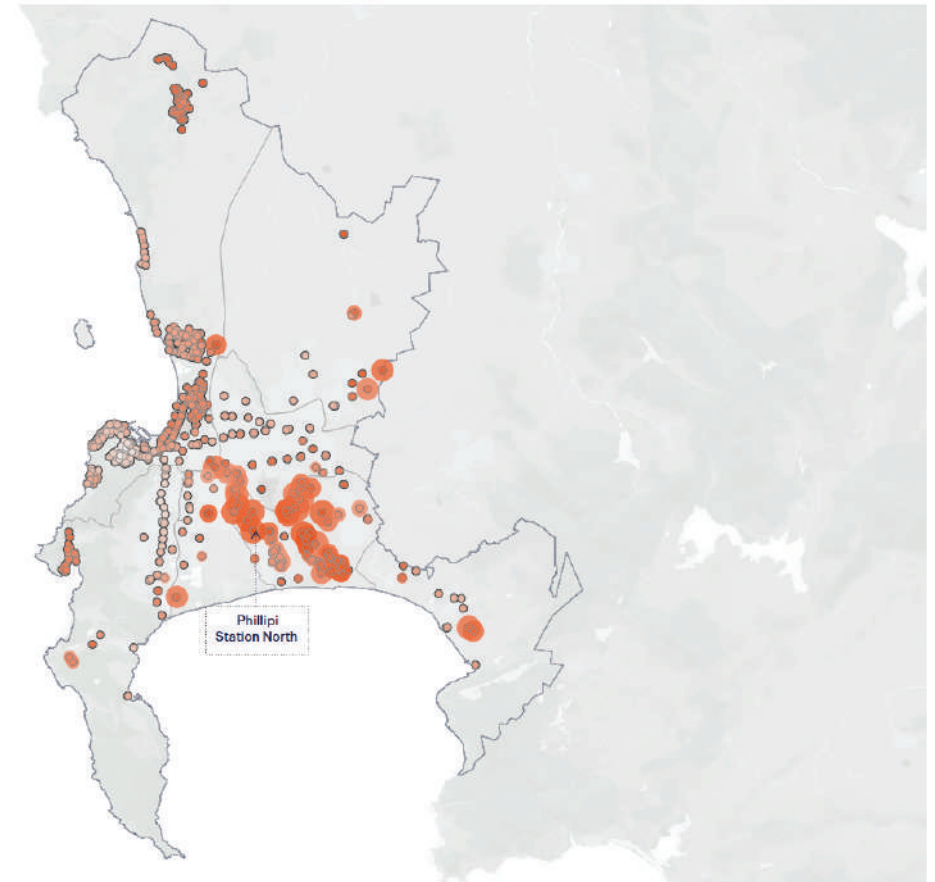
Phillipi Station North Performance



LaDataViz

© 2025 Mapbox © OpenStreetMap

Total Score 0.000



1.000

5.2 Development of the Key Performance Indicators

Defining "opportunity" in measurable terms required developing a framework that captures multiple dimensions of urban wellbeing while remaining tractable for systematic analysis. The research drew on foundations described in section 4 to identify five Key Performance Indicator categories representing the essential services and resources that enable economic and social participation:

Basic Services

encompasses foundational infrastructure determining whether households can meet fundamental needs: informality (households in backyard or squatter shacks), toilet insecurity (households without toilets or using bucket/ecological systems), energy insecurity for heating and lighting (households dependent on paraffin, wood, coal, candles, or gas), piped water insecurity (households without in-home access or dependent on communal points), and healthcare accessibility (street network distance to hospitals within 5km).

Learning and Skills

addresses human capital dimensions, measuring the youth population with education beyond high school, street accessibility to schools within 5km, and youth population density (ages 15-24). High youth concentrations indicate areas where learning interventions could achieve substantial impact.

Work and Enterprise

captures economic opportunity through unemployment ratios among the working-age population, median income levels, and household dependency ratios (the proportion of elderly residents relying on younger earners' income).

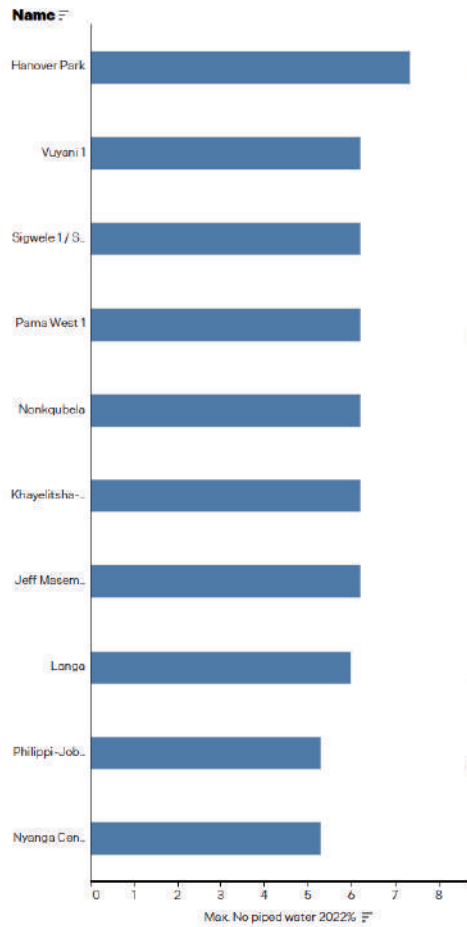
Digital and Media Access

measures connectivity through household internet access, cell phone ownership, and computer availability. These metrics identify areas where residents are excluded from online services, remote work opportunities, and digital literacy development.

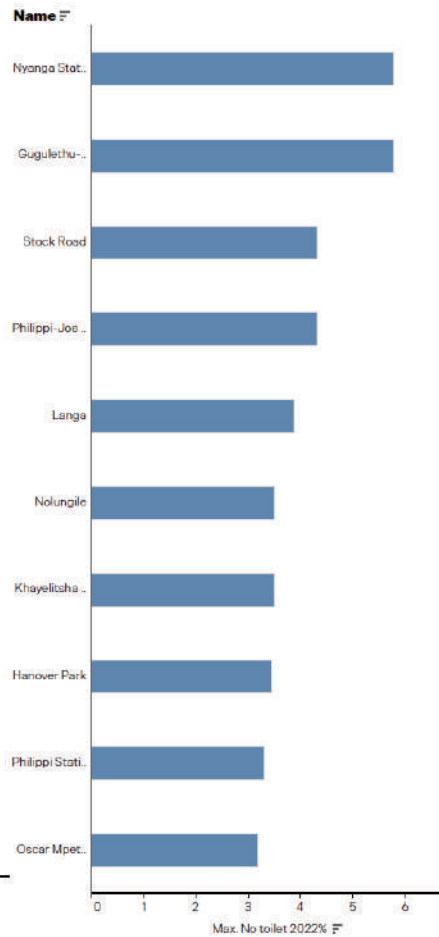
Civic and Administration

assesses access to government services through street network distance to libraries, police stations, municipal offices, and post offices within 5km. Administrative exclusion compounds other forms of disadvantage; residents unable to access civic services face barriers to documentation, licensing, and formal economic participation.

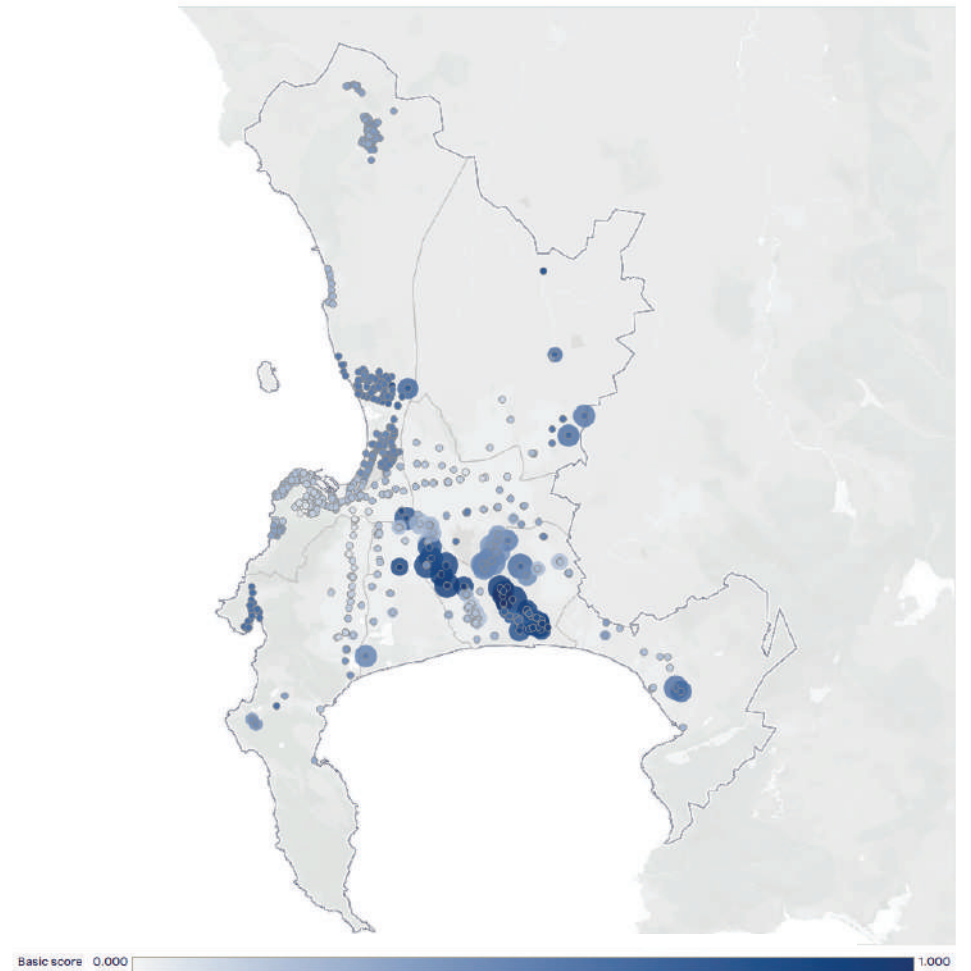
% of Households without piped Water



% of Households without Sanitation



Basic Service Performance



☒ Informality
 ☒ Water Insecurity
 ☒ Sanitation
 ☒ Light Energy

Dashboard for Basic Services

Source © [Cape Town] Cohort, Norman Foster Institute. 2025

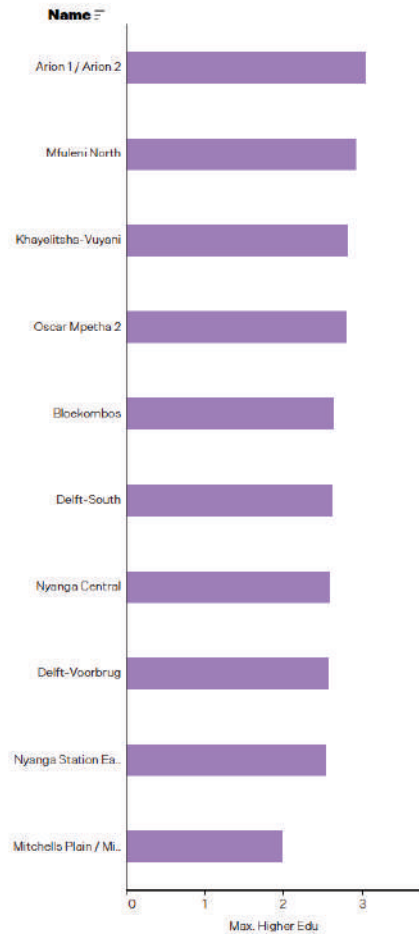
5.3 Data Sources and Analytical Methods

Populating the 19 metrics across 610 nodes required integrating multiple data sources: the 2011 and 2022 South African Censuses for demographic and household infrastructure data, the 2024 City of Cape Town Household Survey for transport user volume estimates, The Municipalities Department of Parks and Recreation for granular 2020 and 2040 population data, OpenStreetMap for spatial accessibility calculations, the Cape Town Municipality Open Data Portal for service facility locations, and the Transport Network Plan 2032 for commuter route data identifying the most heavily used routes into the Central Business District.

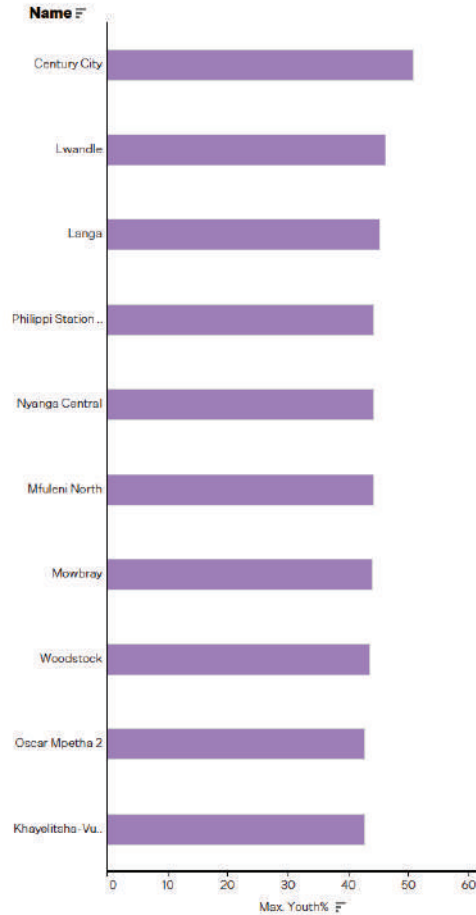
The analytical challenge lay in assigning data collected at different spatial units (pixels, census tracts, suburbs, wards, and police precincts) to point-based transport nodes. The team addressed this through spatial joins, assigning each node the characteristics of its containing spatial unit. For example, the Langa taxi rank received the Basic Services data associated with the Langa suburb, prioritizing smaller spatial units where available for greater precision.

All metric values were normalized to a min-max 0-1 scale using an internal ranking of the 610 nodes analysed. Scores indicate relative need within Cape Town's transport network rather than comparison against external standards: 0 represents the best-performing node for that metric, and 1 represents the node with the greatest need. This approach enables meaningful comparison across metrics, while acknowledging that prioritization occurs within Cape Town's specific urban context.

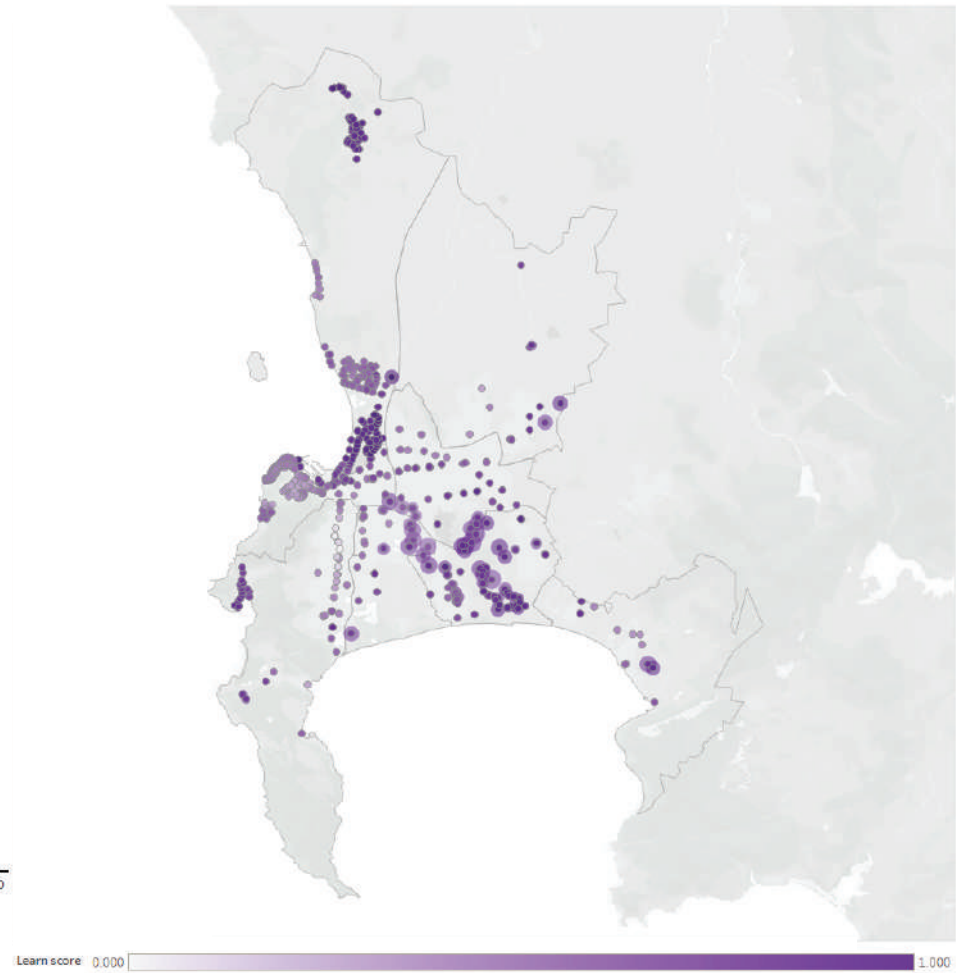
Lowest % of youth with Higher Education



% Youth Population



Learning and Skills Performance



5.4 The Geography of Opportunity Deficit

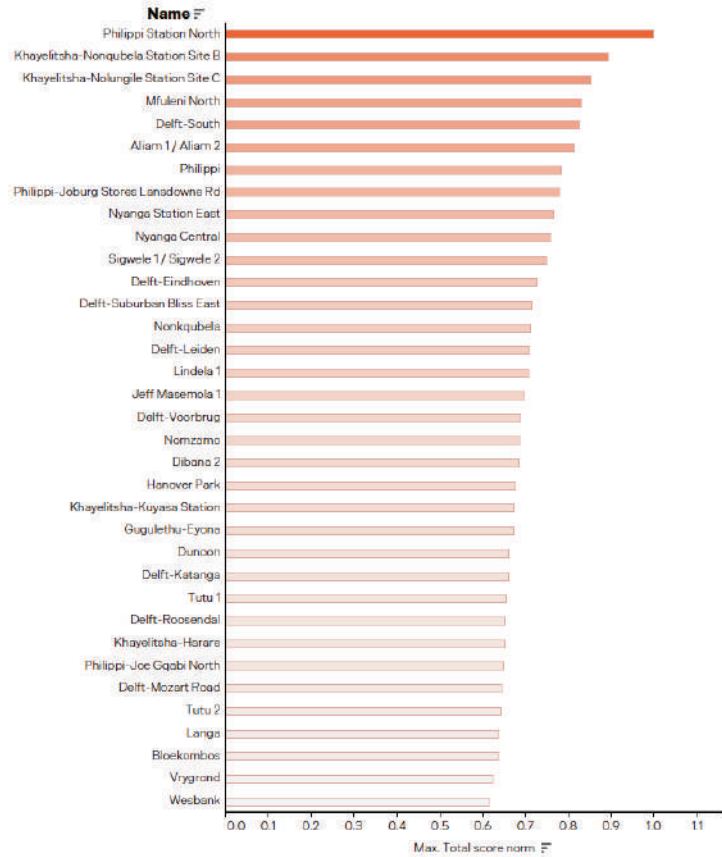
The 610-node analysis revealed stark spatial patterns confirming apartheid's persistent legacy. Nodes with the highest aggregate opportunity deficits concentrate in the southeastern townships and Cape Flats, with the top-ranked priority nodes in Philippi, Khayelitsha, Delft, and Nyanga. Philippi Station North emerged as the highest-priority (normalized score of 1.0000), followed by Khayelitsha-Nonqubela Station Site B (0.8932), Khayelitsha-Nolungile Station Site C (0.8532), and Mfuleni North (0.8287). The majority of the 35 highest-priority nodes fall within areas historically designated for Black and Coloured residents under the Group Areas Act, demonstrating that spatial inequality continues to shape opportunity decades after its repeal.

Beyond historically marginalized areas, the analysis identified four priority nodes near the Urban Development Boundary. Dunoon, Bloekombos, Nomzamo, and Vrygrond all ranked within the top 35. This indicates not only the legacy of racial segregation but also contemporary patterns of spatial exclusion affecting communities at the metropolitan periphery.

A notable finding concerning node classification: Philippi Station North, despite being classified as a minor minibuss taxi, emerged as the highest-priority site. This reflects the tool's capacity to identify locations where modest transport infrastructure serves populations experiencing acute, multidimensional deficits, which is where Opportunity Hub interventions could achieve transformative impact.



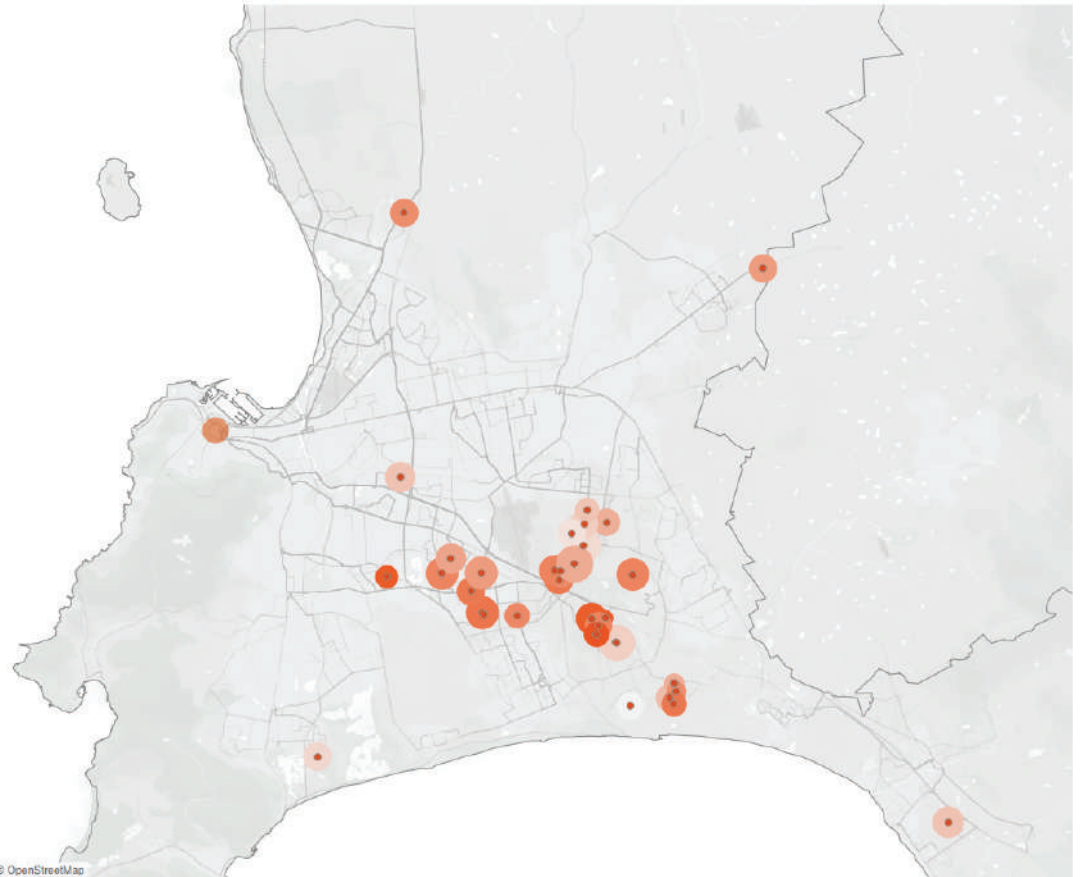
Top 35 Priority Nodes



Total Opportunity Score

0.7607

Top 35 Priority Nodes in Capetown



© 2025 Mapbox © OpenStreetMap

1.0000

Dashboard showing the Top 35 Priority Nodes

Source © [Cape Town] Cohort, Norman Foster Institute. 2025

Basic Service deficits concentrate in nodes within the Mitchells Plain/Khayelitsha district and across the Cape Flats, specifically in suburbs characterized by informal settlements. The analysis revealed a strong correlation between informality and basic services deficits: areas with high concentrations of backyard dwellings and squatter settlements consistently exhibited elevated need across water, sanitation, energy, and healthcare accessibility metrics. Notably, Langa nodes emerged as experiencing significant basic services deficits despite the suburb's location within the Table Bay district, proximate to the Central Business District. This finding underscores that geographic proximity to economic centres does not automatically translate to service provision, particularly in historically Black townships that experienced systematic underinvestment during the apartheid era.

Learning and Skills analysis revealed differentiated patterns. The greatest overall need for learning and skills interventions concentrates in nodes within the Blaauwberg District, a finding driven primarily by low accessibility to schools within 5 kilometres. However, when examining higher education attainment specifically, a different geography emerges: Mitchells Plain, Nyanga, and Delft nodes demonstrate the most acute need for interventions addressing post-secondary education and advanced skills development. These patterns suggest that while school accessibility challenges affect the Blaauwberg periphery, the concentration of residents without higher education qualifications centres on the established township areas of the Cape Flats, indicating distinct intervention requirements

across the learning spectrum.

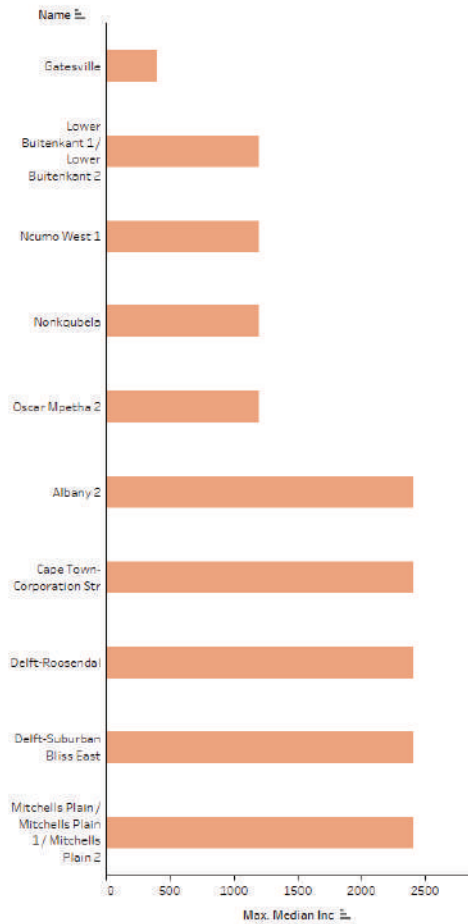
Work and Enterprise are most pronounced in nodes within Delft, Nyanga, and Gugulethu suburbs. This finding carries particular historical significance: these three areas were planned as townships under the apartheid government, with Nyanga and Gugulethu designated for the Black population and Delft for the mixed-race population classified as "Coloured" under apartheid racial categories. The persistence of work and enterprise deficits in these locations reflects the enduring economic consequences of forced removals and spatial planning designed to exclude non-white residents from economic opportunity. The correlation between historical township status, contemporary informality, and work/enterprise deficits underscores the need for interventions that create pathways to formal economic participation.

Digital and Media exclusion concentrates in nodes within the Mitchells Plain/Khayelitsha district, the southern Tygerberg area around Delft, and the northern regions of Blaauwberg. Unsurprisingly, residents proximate to the Central Business District and the Southern District close to Table Bay emerged as the most digitally connected populations in the metropolitan region. The concentration of digital deficits in peripheral areas compounds other forms of disadvantage, as residents lacking internet access, computers, and in some cases mobile phones face exclusion from increasingly digitized government services, employment platforms, educational resources, and financial services. This data suggests that digital access interventions

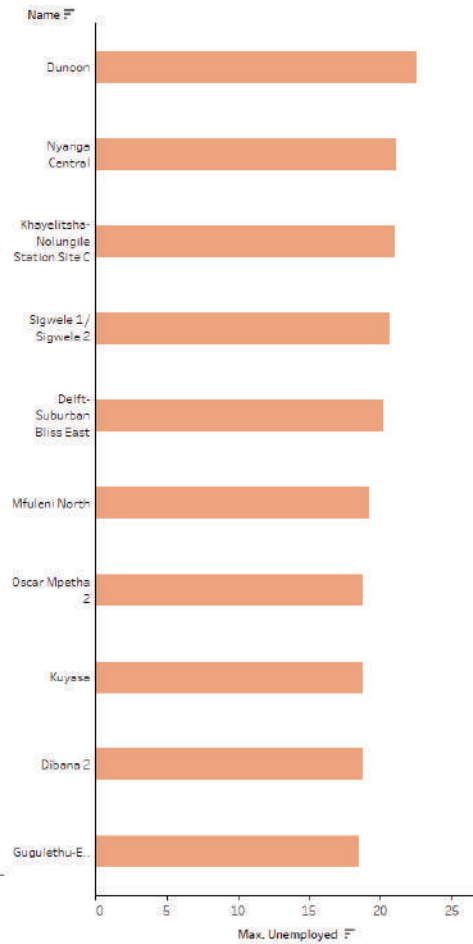
could serve as multipliers, enhancing residents' capacity to address basic services, learning, and employment needs through online channels.

Civic and Administration services concentrate in the Table Bay District within the Central Business District and along the railway corridor in the Southern District. This spatial distribution creates pronounced accessibility challenges for residents in peripheral areas, with nodes in the northern part of Blaauwberg district emerging as most in need of administration services. The Blaauwberg finding likely reflects the area's position near the boundary between the City of Cape Town Municipality and the West Coast District Municipality, creating a service gap where residents fall between administrative jurisdictions. Administrative exclusion in these areas means that residents face greater distances and travel costs to access documentation services, municipal offices, libraries, and police stations, barriers that compound economic disadvantage by increasing the difficulty of formal sector participation, business registration, and civic engagement.

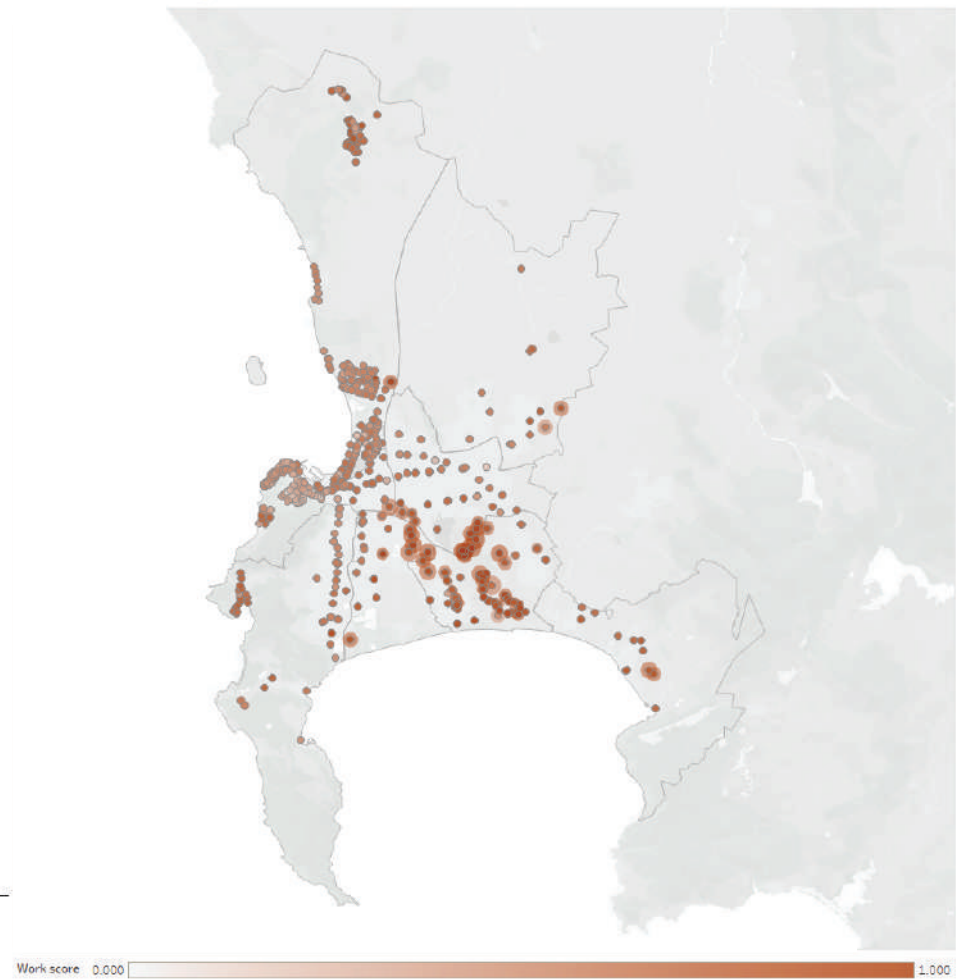
Low Median Income in Rand



% of Unemployment



Work and Enterprise Performance



☰ Dependency Ratio ✕ Median Income

Dashboard for Work & Enterprise

Source © [Cape Town] Cohort, Norman Foster Institute. 2025

5.5 Priority Node Identification

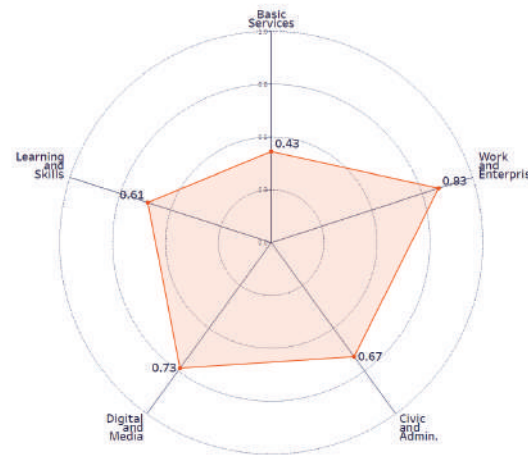
The final analytical step applied a weighting system to identify nodes where intervention would address both high need and high impact. Raw KPI scores were weighted by three factors: estimated transport user volumes at each node derived from the 2024 Household Survey, current population density (2020), and projected population growth (2040). This weighting ensures that prioritization captures not only the severity of opportunity deficits but also the potential reach of interventions.

The weighted analysis generated a ranked list of priority nodes, with the top 35 representing sites where pilot interventions could demonstrate proof of concept while addressing the most acute needs. These nodes concentrate predominantly in the southeastern townships of Philippi, Khayelitsha, Delft, and Nyanga, with additional priority nodes in Gugulethu, Mitchells Plain, and Langa. The geographic distribution confirms that apartheid-era spatial planning continues to determine where opportunity deficits concentrate, while the inclusion of peripheral outliers at Dunoon, Bloekombos, Nomzamo, and Vrygrond demonstrates that the framework also captures contemporary patterns of geographic marginalization affecting communities at the urban edge.

CBD



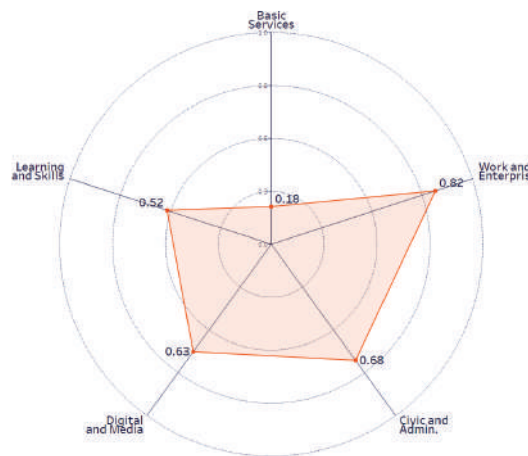
CBD Commuter



Langa



Mitchells Plain



5.6 Node Profiles: Diagnostic Precision

Beyond aggregate prioritization, the analytical framework enables detailed diagnostic profiles for individual nodes. These profiles disaggregate composite scores to reveal precisely which metrics drive a node's priority status, thereby informing programmatic decisions. Radar chart visualizations present normalized values across all 19 metrics, allowing planners to identify which specific interventions the catalogue should emphasize.

The Central Business District nodes illustrate the framework's diagnostic sophistication. If the analysis relied solely on census data for the approximately 7,000 CBD residents, these nodes would appear low-need across most indicators, as the formal residential population enjoys high service access. However, the team recognized that CBD transport nodes serve a fundamentally different population: the daily commuters who flow through these locations en route to employment. Drawing on the Transport Network Plan 2032, the team identified the most used commuter routes into the CBD, which originate primarily from Atlantis, Mitchells Plain, and Nyanga. The CBD node profiles were therefore constructed using demographic and socioeconomic data from these commuter origin areas rather than CBD resident data. This methodological innovation ensures that Opportunity Hub programming at CBD nodes responds to the actual populations passing through, working-class commuters from peripheral townships, rather than the relatively affluent formal residents. The resulting profiles show CBD nodes with elevated need across Work and Enterprise and Learning and Skills dimensions, accurately reflecting the commuter population's characteristics.

Top Left: Central Business District; Top Right: CBD Commuter; Bottom Left: Langa; Bottom Right: Mitchells Plain

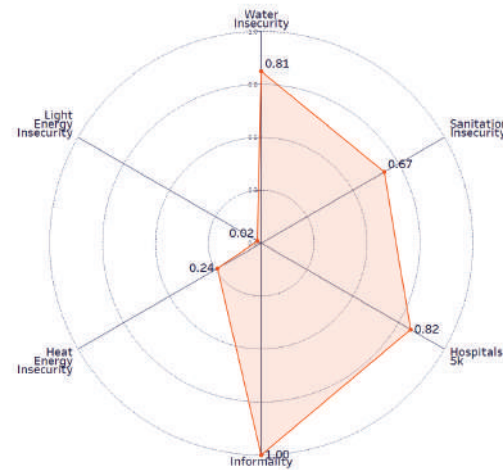
Source © [Cape Town] Cohort, Norman Foster Institute. 2025

The Langa taxi rank further illustrates this diagnostic capability.

The node's profile reveals a Learning and Skills score of 0.87, indicating very low rates of higher education attainment in the surrounding population, a finding that directly suggests educational and skills programming as intervention priorities.

Simultaneously, Langa's Basic Services score reflects the suburb's challenges with informality and infrastructure despite its proximity to the CBD, while Work and Enterprise metrics capture the economic marginalization that persists in this historically Black township. This granular diagnostic capacity ensures that interventions respond to demonstrated local needs rather than applying generic solutions. A node with high Digital and Media deficits but moderate Basic Services scores would receive different catalogue recommendations than a node with the inverse profile.

Basic Services



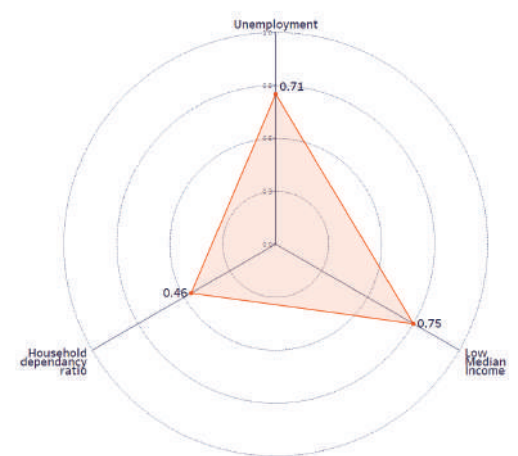
Learning & Skills



Digital & Media



Work & Enterprise



Top Left: Basic Service Langa; Top Right: Learning & Skills Langa; Bottom Left: Digital & Media Langa; Bottom Right: Work & Enterprise Langa
Source © [Cape Town] Cohort, Norman Foster Institute. 2025

5.7 From Analysis to Design Requirements

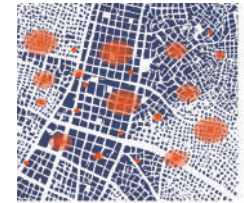
The research findings generated several design requirements shaping the methodology presented in Section 6. The concentration of opportunity deficits at transport nodes validated the intervention typology while establishing that hubs must accommodate transient, time-constrained commuter populations. Interventions must deliver value within dwell times ranging from 5 minutes at quick-service stops to 30+ minutes at major interchanges.

The diversity of deficit profiles across nodes required a flexible design system capable of responding to varied programmatic requirements. This finding directly informed the catalogue matrix approach, which matches scale and dwell time with KPI-specific programming.

The correlation between opportunity deficits and population concentrations in peripheral townships confirmed the strategic value of interventions replicable at scale rather than designed as singular flagship projects. The methodology therefore emphasizes standardized structural systems that enable rapid deployment while accommodating local customization.

Finally, the research revealed that the 610-node network far exceeds feasible short-term implementation capacity. The phased approach (4 pilots in the first year, 25 hubs in the first 10 years, and 120 by 2050) reflects this realistic assessment of what the evidence base can support.

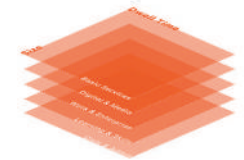
1. Prioritization



2. Site Selection



3. Catalogue



4. Participation



5. Intervention



Method explained in 5 steps

Source © [Cape Town] Cohort, Norman Foster Institute. 2025

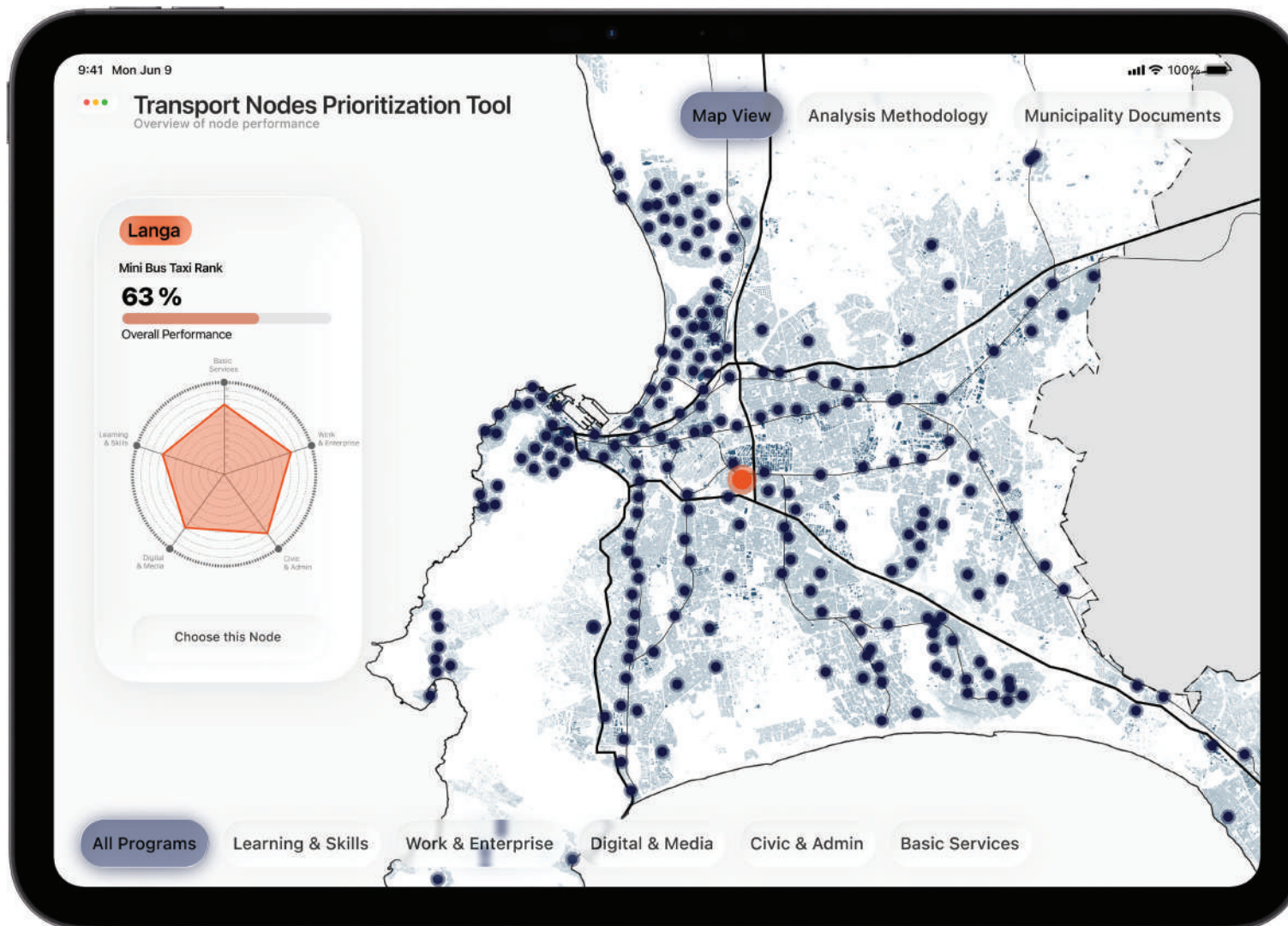
6. Method

6.1 Prioritization Through Data-Driven Analysis

The prioritization stage employs the Opportunity Explorer tool, an analytical framework that evaluates all 610 public transport nodes across Cape Town through five Key Performance Indicators (KPIs): Basic Services, Learning and Skills, Work and Enterprise, Digital and Media Access, and Civic and Administration. Drawing from the 2011 and 2022 South African Censuses, the 2024 City of Cape Town Household Survey, The Municipalities Department of Parks and Recreation, OpenStreetMap, and the Integrated Public Transport Network Plan 2032, the tool aggregates 19 distinct metrics to measure opportunity deficits at each node.

Normalized scores (0-1, with higher values indicating greater need) are weighted by three factors: estimated transport user volumes, 2020 population density, and projected 2040 population growth. This ensures prioritization captures both immediate need and long-term impact potential, identifying the top 35 nodes with high opportunity deficits and significant population impact.

Critically, KPIs translate into programmatic interventions. When a node scores highly for lack of basic services (0.87 in Langa's case), this suggests specific modules from the catalogue focused on water access, sanitation, and energy provision. This direct translation from analysis to design recommendation ensures that interventions precisely target identified needs rather than applying generic solutions.



Prioritization Tool Interface, highlighting the Langa Node
Source © [Cape Town] Cohort, Norman Foster Institute. 2025

6.2 Prioritization Through Data-Driven Analysis

Site selection operates at the neighborhood scale, identifying specific intervention locations within a 400-meter radius of prioritized transport nodes. This stage applies the "last mile principle," which recognizes that effective interventions must meet people where they naturally flow rather than requiring additional travel to new destinations. The principle acknowledges that township residents already spend excessive time and resources on transport; interventions should therefore capitalize on existing movement patterns rather than creating new travel demands.

The selection process considers multiple factors beyond pure proximity. Local Spatial Development Frameworks (LSDFs) provide planning context, while existing public space availability determines physical feasibility. The methodology prioritizes spaces that already function as informal gathering points or waiting areas, recognizing the latent social infrastructure present in these locations. In the CBD case study, for example, the selection of Station Deck responds to existing pedestrian flows between the bus terminal and city center, while the Langa site leverages an underutilized public space adjacent to the taxi rank where commuters already congregate.

This approach ensures that Opportunity Hubs integrate with the existing urban fabric rather than imposing new spatial logics. The methodology explicitly avoids creating isolated intervention zones, instead embedding services within the natural circulation patterns of daily life.



Site Selection Tool showing Washington Street in Langa and the possible sites
Source © [Cape Town] Cohort, Norman Foster Institute. 2025

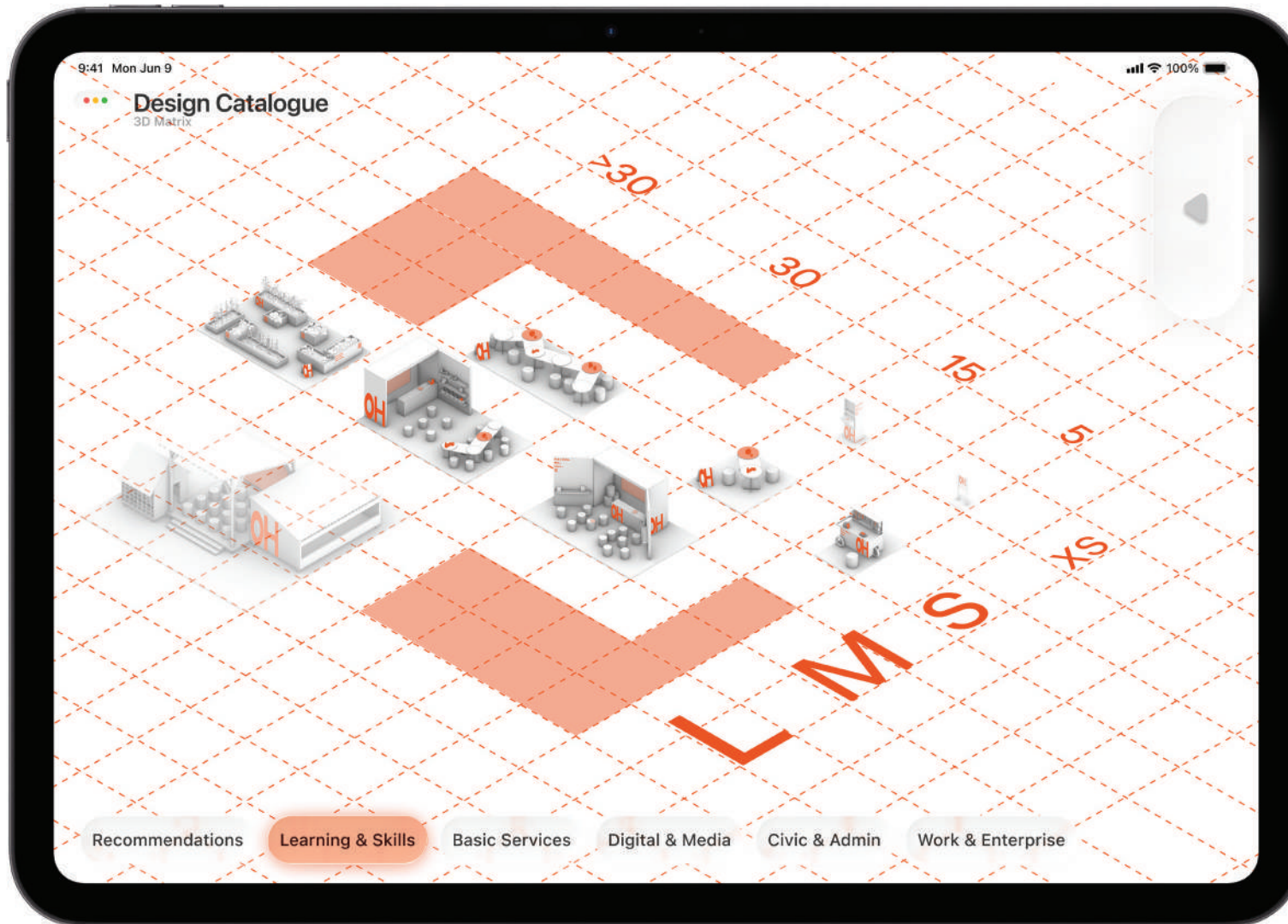
6.3 The Catalogue Matrix System

The catalogue represents the methodology's most innovative contribution to bridging analysis and design. This sophisticated matrix system crosses spatial dimensions (site sizes) with temporal dimensions (dwell times) to generate appropriate intervention types. The matrix operates across four size categories: extra small ($\leq 10\text{m}^2$), small ($10\text{--}50\text{m}^2$), medium ($50\text{--}100\text{m}^2$), and large ($>100\text{m}^2$), each matched with four dwell time categories: 5 minutes, 15 minutes, 30 minutes, and more than 30 minutes.

Every size category includes common program elements that form the infrastructural backbone of all interventions: energy provision, connectivity, lighting, shade, and rest facilities. Medium and large interventions additionally incorporate sanitation facilities, recognizing that spatial thresholds enable more complex services. These common programs ensure that all Opportunity Hubs, regardless of scale, provide essential urban amenities currently lacking in township environments.

The specific program layer responds directly to the KPI analysis from the prioritization stage. Each of the five KPI categories translates into a distinct program matrix, populated with targeted interventions. For Learning and Skills, extra small interventions with 5-minute dwell times might include repair points that facilitate "learning through repairing," while medium-scale 30-minute interventions could accommodate maker spaces for "learning through practicing." This granular matching of program, scale, and time ensures that interventions respond to both spatial constraints and user behavior patterns.

Research validates this approach to modular design, demonstrating that standardized components can effectively enable customization. As Lidelöw (2024) notes in an analysis of modular construction systems, the balance between standardization and flexibility is achieved through creating "a number of standardized modules in a catalogue" that can be configured to meet specific client needs. The matrix system operationalizes this principle, providing sufficient standardization for efficient production while maintaining flexibility for local adaptation.



Catalogue Matrix System showing the Learning & Skills Matrix for the Langa Opportunity Hub
Source © [Cape Town] Cohort, Norman Foster Institute. 2025

6.4 Community Participation and Co-Design

The participation stage transforms the technical catalogue into a collaborative design process. Rather than presenting communities with predetermined solutions or unlimited options, the methodology provides a curated set of possibilities filtered through the prioritization analysis. This approach respects community agency while ensuring technical feasibility and resource efficiency.

Community engagement sessions utilize the catalogue matrices as "igniters of conversation," presenting visual representations of possible interventions scaled and programmed according to local conditions. The pre-selection based on KPI analysis ensures that proposed options directly address identified needs while remaining open to community refinement and prioritization. For instance, if the prioritization tool identifies high scores for both Basic Services and Learning and Skills, the engagement process might present various combinations of water access points, sanitation facilities, and educational spaces, allowing the community to determine the specific configuration that best serves their needs.

This filtered choice model acknowledges that effective participation requires structure without prescription. Communities possess intimate knowledge of local dynamics that data cannot capture, which can fundamentally shape the success of interventions. The method positions communities as co-designers rather than consultants, with genuine influence over final configurations while working within technically viable parameters.



Community Engagement Workshop, Gugu S'thebe, Langa
Source © [Cape Town] Cohort, Norman Foster Institute. 2025

6.5 Intervention Through Modular Construction

The physical intervention employs cross-laminated timber (CLT) modular construction, utilizing standardized structural frames that enable both immediate implementation and long-term adaptability. The system comprises three frame modules (3x3m, 3x4m, and 3x5m) constructed from locally sourced CLT, with beveled joints enabling multiple configuration possibilities. This approach exemplifies how "modular construction combines standardized core elements with bespoke, customizable features, enabling companies to meet specific client needs without undermining operational efficiency"⁴⁴.

The selection of CLT as the primary structural material responds to multiple imperatives. Environmental research demonstrates approximately a 40% reduction in carbon footprint when using CLT instead of conventional construction materials for multi-story buildings⁴⁵. More significantly in Cape Town's context, CLT production can utilize local eucalyptus wood, an abundant, non-indigenous species, creating circular-economy opportunities. This aligns with Cape Town's existing CLT initiatives, such as the XLAM project in Langa, which demonstrates the viability of local CLT production using regional timber resources.

The modular system deliberately limits prescription to structural frames, leaving cladding, roofing, and infill elements open for local determination. This approach serves multiple purposes: it enables local employment in construction completion, allows for material choices responsive to specific site conditions, and facilitates incremental upgrading as resources become available.

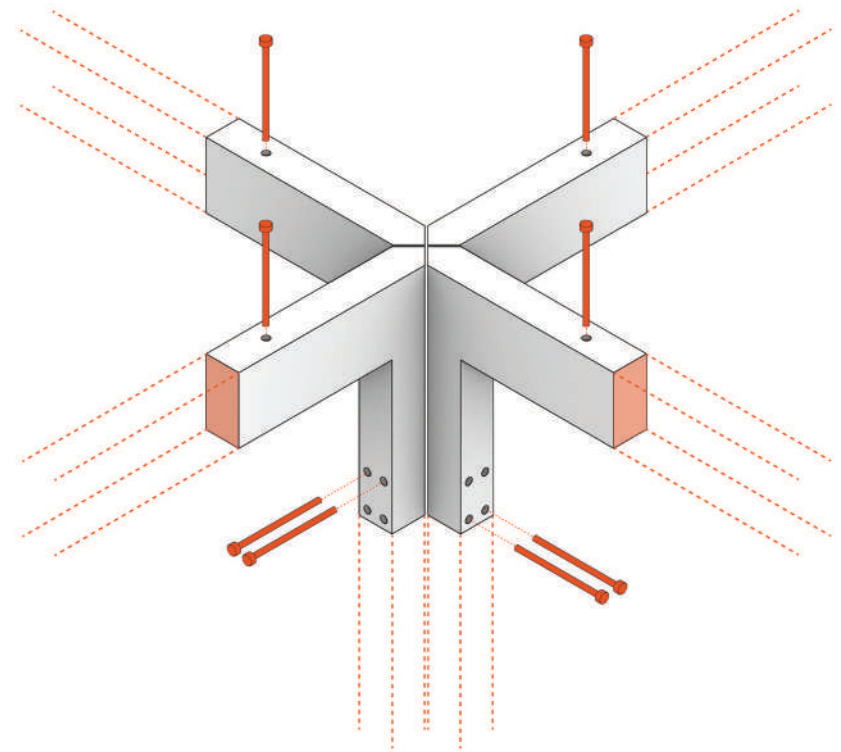
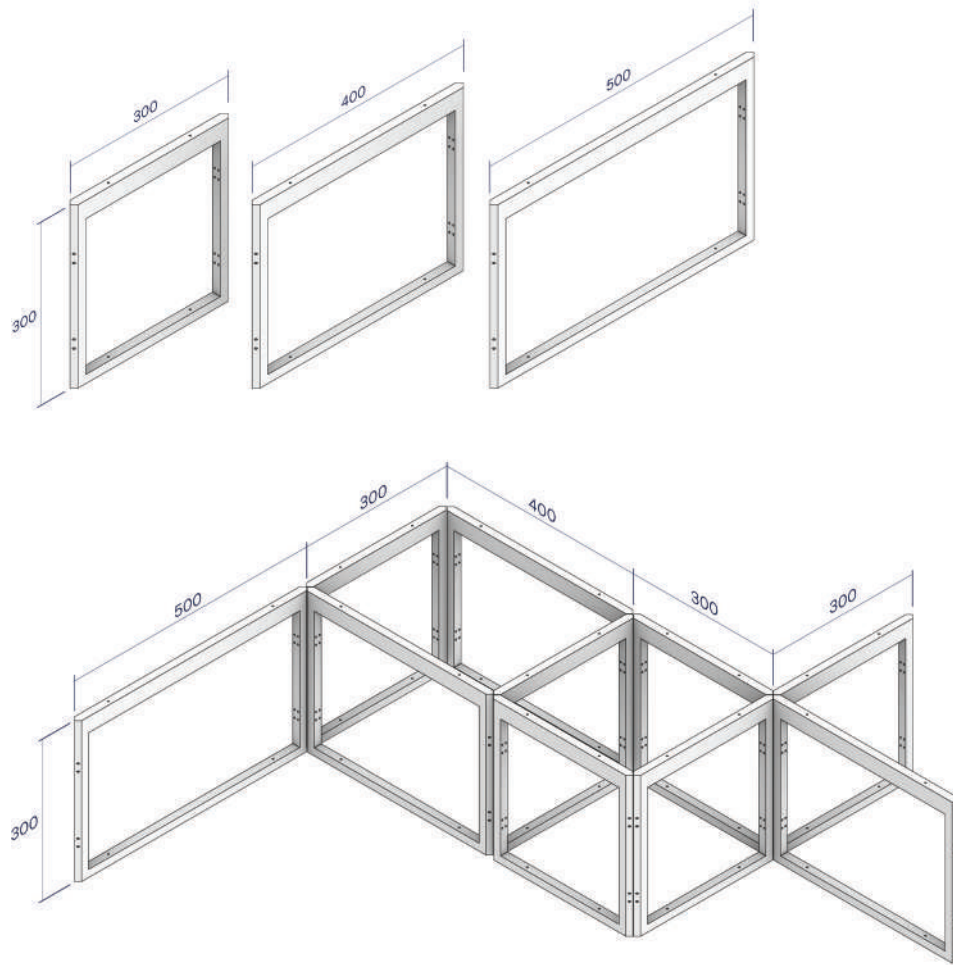
The frames provide the essential structural integrity and dimensional standardization necessary for systematic deployment, while the remaining surfaces offer opportunities for community expression and adaptation.

Critically, the methodology embraces "intentional obsolescence" as a success metric. The progression sequences demonstrated in the Langa and CBD case studies illustrate how Opportunity Hubs can catalyze broader urban transformation. Initial frame installations provide immediate services while establishing spatial claims for public amenity. As usage intensifies and value is demonstrated, municipal investment in permanent infrastructure becomes politically and economically viable. The modular frames can then be disassembled and redeployed elsewhere, with mechanical fastening systems enabling "high-value Mass Timber elements to be disassembled and then reused or recycled to be endlessly recirculated"⁴⁶. The ultimate success manifests not in permanent Opportunity Hubs, but in dignified public spaces and integrated urban services that render the temporary interventions unnecessary.

⁴⁴ Pit, 2024

⁴⁵ Younis & Doodoo, 2022

⁴⁶ Ross et al., 2023



Modular System Approach, showing the Measurements, the Joint & the System itself
 Source © [Cape Town] Cohort, Norman Foster Institute. 2025

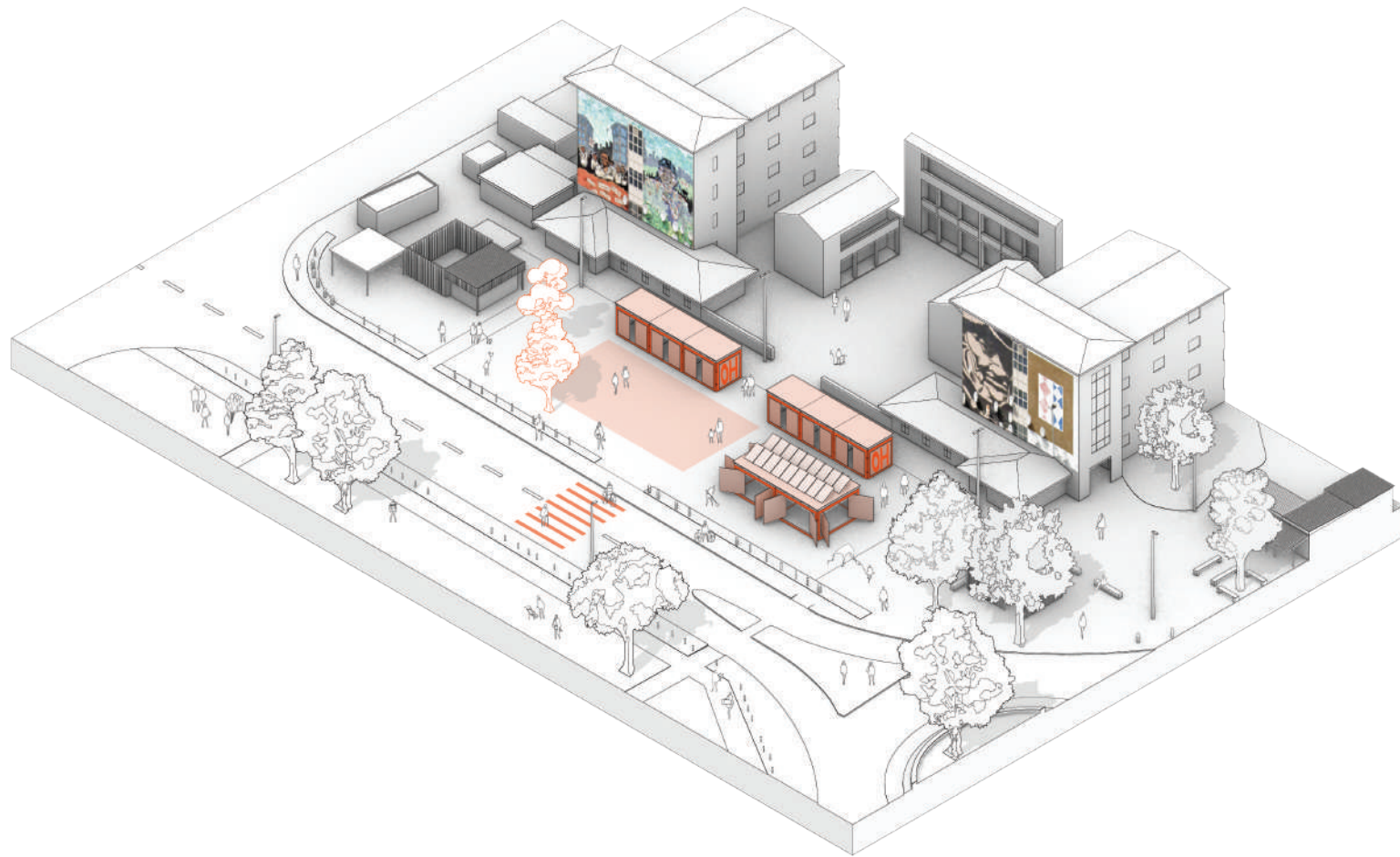
6.6 Integration and Scalability

The five-stage method creates a replicable yet adaptable framework for citywide implementation. The standardization embedded in the prioritization tool, catalogue matrix, and modular construction system enables scaling. Simultaneously, the community participation stage and open-ended material specifications ensure that each intervention responds to local specificity.

This balance between standardization and customization addresses a challenge in urban intervention: how to achieve the efficiency necessary for large-scale implementation while maintaining the contextual sensitivity. The methodology demonstrates that these objectives need not be at odds; rather, strategic standardization at structural levels enables meaningful customization at the human scale. As research on modular systems confirms, "taking basic units conforming to modular network scales provides prerequisites for improving standardization rates while combination modes with different quantities, depths, and splicing methods guarantee diversity at the macro level"⁴⁷.

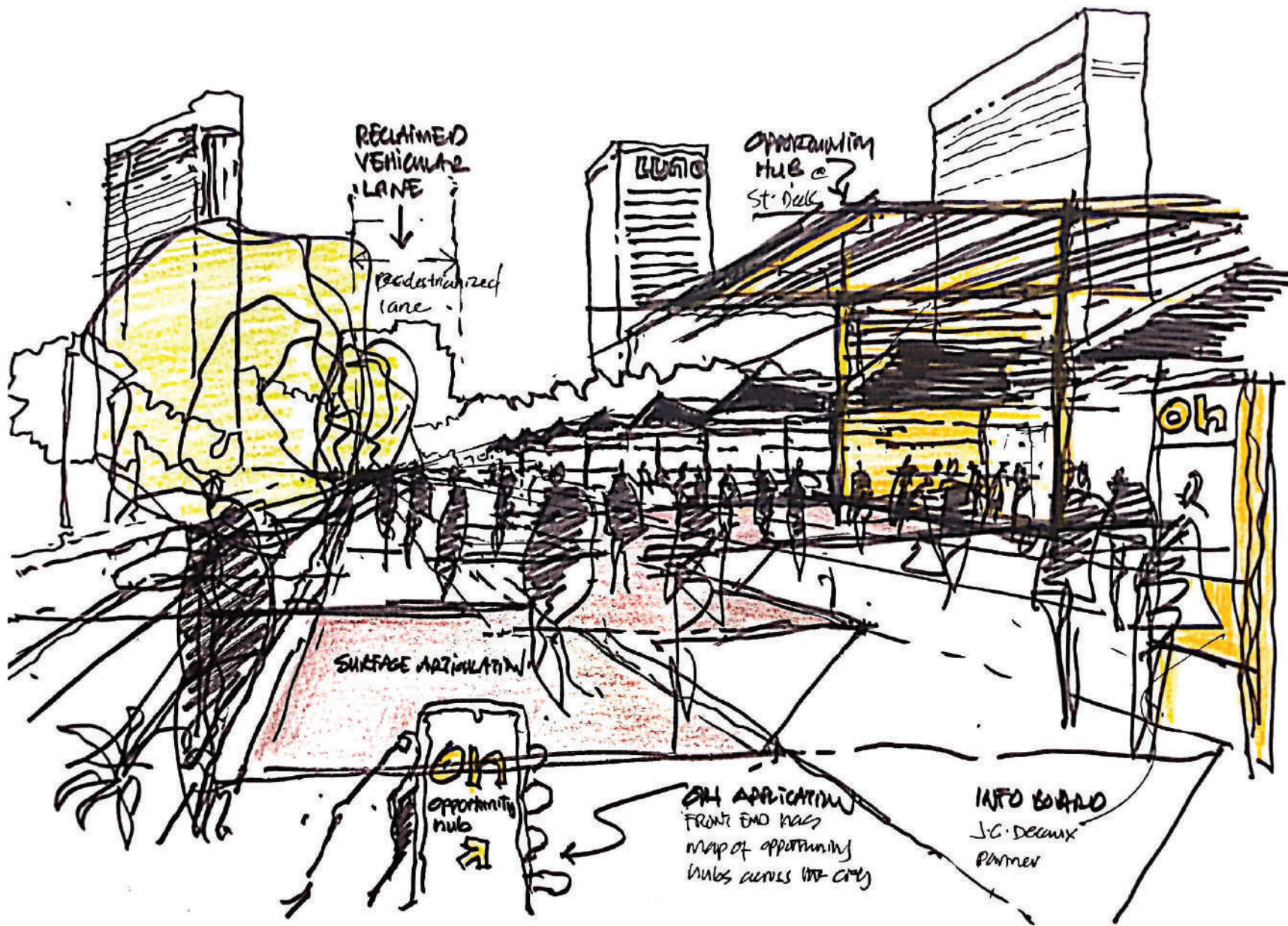
The methodology's innovation lies not in any single component but in their systematic integration. Data analysis informs but doesn't determine design. Community knowledge supplements but doesn't override technical requirements. Standardized elements enable rather than constrain local adaptation. This careful calibration positions Opportunity Hubs as catalysts for broader transformation, creating the spatial and social infrastructure necessary for township communities to access the opportunities that spatial apartheid continues to deny.

⁴⁷ Zhang et al., 2024



Axonometric Drawing of Opportunity Hub in Washington Street, Langa
Source © [Cape Town] Cohort, Norman Foster Institute. 2025

7. Case Studies



Sketch of the Design Process of an Opportunity Hub
 Source © [Cape Town] Cohort, Norman Foster Institute. 2025

The following case studies demonstrate how the five-stage methodology translates into site-specific interventions across two contrasting urban contexts: Langa, Cape Town's oldest formal township, and the Central Business District surrounding the Station Deck transport interchange. These pilot sites were selected to test the methodology's adaptability across fundamentally different spatial, social, and economic conditions. Langa represents the township typology in which opportunity deficits are most acute, but where community cohesion and existing social infrastructure offer implementation advantages. The CBD site represents a high-throughput environment where interventions must respond to transient populations and complex stakeholder dynamics. Together, these pilots establish replicable models for the broader network of 120 Opportunity Hubs envisioned by 2050.

7.1 Langa Opportunity Hub

Context and Prioritization

Langa's selection as a primary pilot site reflects both analytical findings and practical considerations. As Cape Town's first formal township, established in 1927, Langa carries profound historical significance while demonstrating the enduring spatial patterns of segregation that the Opportunity Hub intervention seeks to address. The Opportunity Explorer tool identifies Langa's primary transport node, the taxi rank on Washington Street, as ranking within the top 35 priority nodes citywide, with particularly acute deficits in Basic Services (0.73), Learning and Skills (0.69), and Digital and Media Access (0.7). These scores indicate that households in the surrounding area experience significant gaps in access to water, sanitation, educational facilities, and digital connectivity.

Washington Street emerged from spatial analysis as Langa's principal mobility corridor, concentrating pedestrian and vehicular flows throughout the day. The taxi rank serves as the neighbourhood's primary interface with the metropolitan transport network, processing thousands of daily commuters travelling to employment centres across Cape Town. This concentration of movement creates precisely the conditions the last-mile principle identifies as optimal for intervention: a location where services can reach the maximum population without requiring additional travel.



Radarchart showing the performance in all programmes for Langa
Source © [Cape Town] Cohort, Norman Foster Institute. 2025



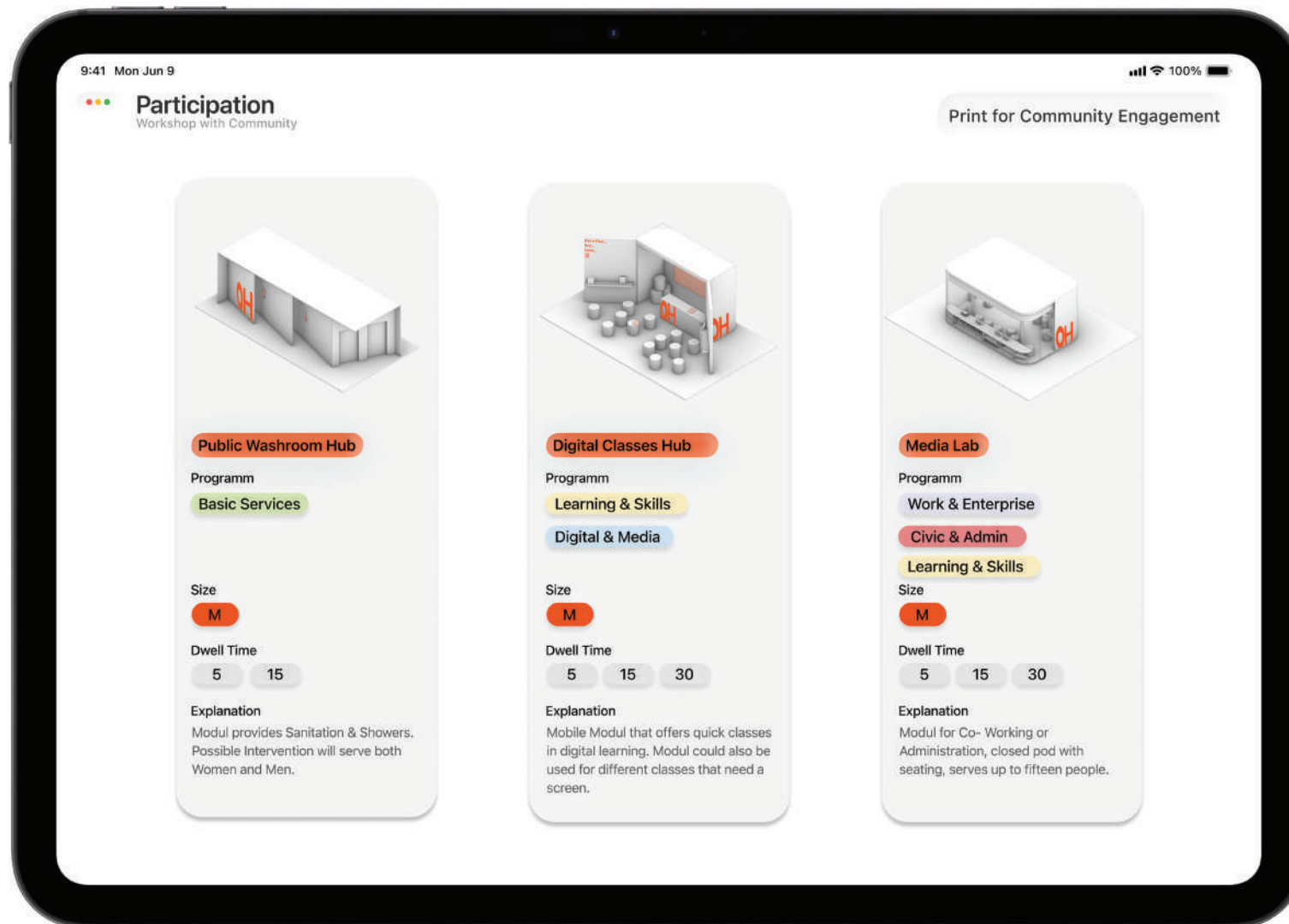
Public Space in Langa

Source © [Cape Town] Cohort, Norman Foster Institute. 2025

Site Selection and Programming

Site selection within the 400-metre catchment identified a publicly owned parcel on the south side of Washington Street, directly adjacent to dense residential blocks near the taxi rank. The site's qualities (high visibility, proximity to transport, and strong integration with daily urban life) align with the methodology's criteria for effective intervention locations. Current conditions reveal an underutilized space encroached by informal parking, with degraded surfaces and limited public amenities, yet it already functions as an informal gathering point for commuters and residents.

The catalogue matrix, filtered through Langa's KPI profile, generates a programmatic configuration emphasizing Basic Services, Learning and Skills, and Digital and Media Access. The proposed hub comprises two small pavilions dedicated to basic services providing sanitation, access to piped water, and public electricity alongside one large pavilion for learning and skills programming. This larger structure accommodates a flexible, open classroom suitable for training, upskilling initiatives, and youth programmes. The interstitial spaces between structures are conceived as dignified public areas supporting informal trading, waiting, meeting, and circulation, the everyday activities that currently occur without adequate spatial provision.



Three suggested Interventions from the Design Catalogue for the CBD
 Source © [Cape Town] Cohort, Norman Foster Institute. 2025

Evolution Scenario

The Langa pilot illustrates the methodology's temporal dimension through a series of potential evolution phases spanning 30–40 years. Initial deployment installs the CLT frame pavilions on the existing surface without additional ground treatment, establishing the spatial claim and initiating service provision. As usage patterns emerge and community value becomes evident, subsequent phases might introduce improved ground finishes and public furniture, transforming the space from a site of passage into a social destination. Further iterations could add programming to a Work and Enterprise pavilion, for instance, and expand the hub's functional scope while strengthening the public space network along Washington Street.

The scenario's longer-term projection envisions the hub growing in concert with neighbourhood densification. The modular CLT system accommodates vertical expansion by adding a second story, maintaining architectural coherence while responding to increased demand. This approach ensures that the intervention remains responsive to changing conditions rather than imposing a fixed endpoint. The scenario demonstrates how incremental investment, catalysed by the initial Opportunity Hub deployment, could progressively dignify Washington Street as a civic corridor with appropriate public lighting, vegetation, shade structures, and pedestrian amenities.



Washington Street, Langa, Opportunity Hub Intervention 2025
Source © [Cape Town] Cohort, Norman Foster Institute. 2025



Washington Street, Langa, Opportunity Hub Intervention 2026
Source © [Cape Town] Cohort, Norman Foster Institute. 2025



Washington Street, Langa, Opportunity Hub Intervention 2030
Source © [Cape Town] Cohort, Norman Foster Institute. 2025



Washington Street, Langa, Opportunity Hub Intervention 2035
Source © [Cape Town] Cohort, Norman Foster Institute. 2025



Washington Street, Langa, Opportunity Hub Intervention 2040
Source © [Cape Town] Cohort, Norman Foster Institute. 2025



Washington Street, Langa, Opportunity Hub Intervention 2050 day
Source © [Cape Town] Cohort, Norman Foster Institute. 2025



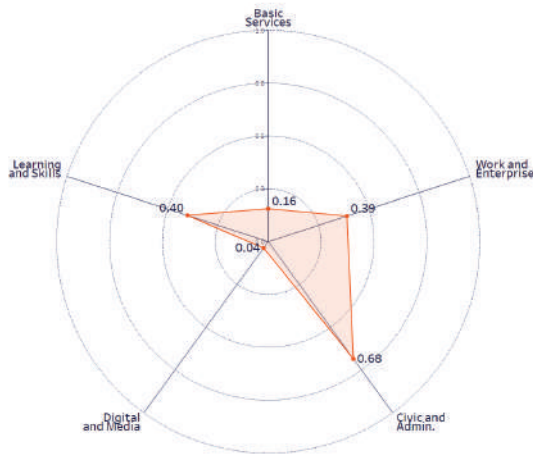
Washington Street, Langa, Opportunity Hub Intervention 2050 night
Source © [Cape Town] Cohort, Norman Foster Institute. 2025

7.2 Central Business District Opportunity Hub

Context and Prioritization

The CBD pilot site presents a fundamentally different intervention context. Analysis of central Cape Town identifies Strand Street as the primary last-mile artery for commuters arriving at the Station Deck and dispersing into the central business district. High pedestrian flows, modal transfers between taxi, bus, and rail services, and intensive informal economic activity concentrate along this corridor. The Station Deck itself represents one of the city's busiest commuter thresholds, yet its immediate surroundings lack the public amenities necessary to support dignified waiting, informal work, or service access.

Unlike Langa, where intervention responds primarily to the needs of the residential population, the CBD site serves a transient population whose presence lasts only minutes rather than hours. The KPI analysis accordingly emphasizes different priorities: Work and Enterprise functions supporting informal vendors and job-seekers, Civic and Administration services enabling commuters to complete administrative tasks during transit, and Digital and Media Access providing connectivity for those navigating the CBD's employment interfaces. This programmatic emphasis reflects the methodology's responsiveness to user behaviour patterns rather than solely to residential demographics.



Top: Radarchart CBD Commuter, Bottom: Radarchart CBD
Source © [Cape Town] Cohort, Norman Foster Institute. 2025

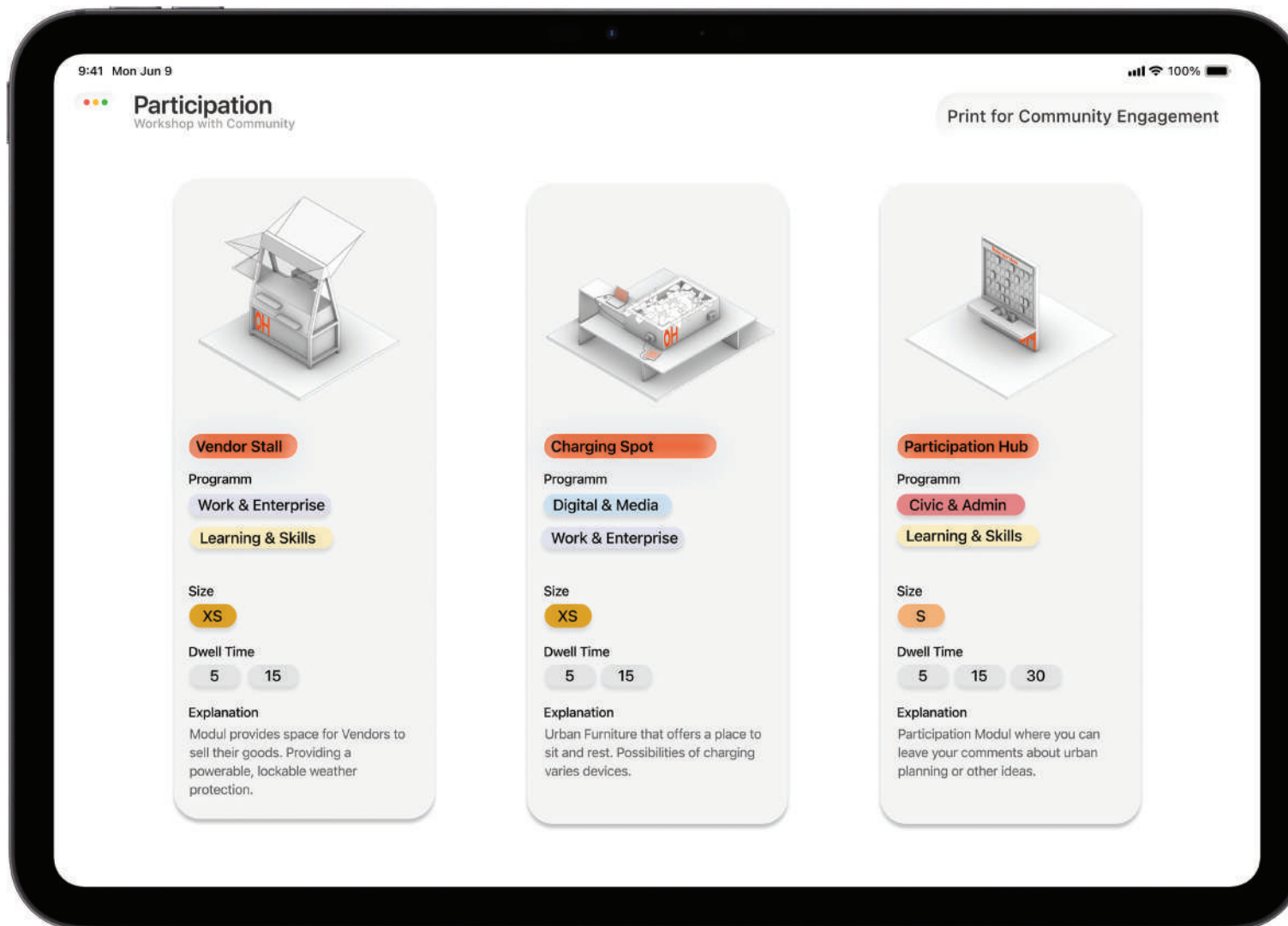


Overview of Central Business District from Table Mountain
Source © [Cape Town] Cohort, Norman Foster Institute. 2025

Intervention Strategy

The CBD intervention employs an initial deployment at the extra-small (XS) scale, recognizing the constrained spatial conditions of sidewalk environments. Small CLT-frame elements placed directly on the existing sidewalk along the Station Deck require minimal infrastructure modification while establishing presence and initiating service provision. This tactical approach enables rapid deployment and low-risk testing of user response before committing to more substantial interventions.

The illustrative evolution scenario for the CBD site demonstrates how successful activation can catalyse broader urban transformation. Increased usage might justify expanding from small frames to semi-enclosed pavilions that offer improved functionality and weather protection. Demonstrated demand could then support municipal action to reclaim adjacent parking lanes for pedestrians, initially through temporary concrete barriers and subsequently through permanent surface treatment. The scenario's culminating phases envision vertical expansion enabling direct connection to the elevated Station Deck taxi rank, integrating the Opportunity Hub into the commuter circulation system itself.



Three suggested Interventions from the Design Catalogue for the CBD
 Source © [Cape Town] Cohort, Norman Foster Institute. 2025



Strand Street, CBD, Opportunity Hub Intervention 2025
Source © [Cape Town] Cohort, Norman Foster Institute. 2025



Strand Street, CBD, Opportunity Hub Intervention 2026
Source © [Cape Town] Cohort, Norman Foster Institute. 2025



Strand Street, CBD, Opportunity Hub Intervention 2027
Source © [Cape Town] Cohort, Norman Foster Institute. 2025



Strand Street, CBD, Opportunity Hub Intervention 2028
Source © [Cape Town] Cohort, Norman Foster Institute. 2025



Strand Street, CBD, Opportunity Hub Intervention 2029 day
Source © [Cape Town] Cohort, Norman Foster Institute. 2025



Strand Street, CBD, Opportunity Hub Intervention 2029 night
Source © [Cape Town] Cohort, Norman Foster Institute. 2025



Strand Street, CBD, Opportunity Hub Intervention 2035
Source © [Cape Town] Cohort, Norman Foster Institute. 2025



Strand Street, CBD, Opportunity Hub Intervention 2040
Source © [Cape Town] Cohort, Norman Foster Institute. 2025

Catalytic Legacy

The CBD scenario articulates a principle central to the Opportunity Hub methodology: these interventions are designed as catalytic rather than permanent installations. Their ultimate success lies in rendering themselves unnecessary as permanent infrastructure emerges. The evolution sequence concludes with the Opportunity Hub's removal, leaving as its legacy a permanently widened pedestrian corridor and dignified public space serving commuters, vendors, and pedestrians along Strand Street. When this transformation occurs, the modular CLT components can be dismantled and redeployed to new priority sites identified through the Opportunity Explorer, ensuring that resources continue to serve areas of greatest need rather than remaining fixed in locations that no longer require temporary intervention.

This catalytic model distinguishes Opportunity Hubs from conventional infrastructure projects. Rather than competing with permanent municipal investment, the hubs demonstrate value and build political support for such investment. They function as spatial placeholders and proof-of-concept installations that reduce the risk associated with larger capital commitments. The CBD pilot, if successful, could set a precedent for similar sidewalk activations along other high-volume pedestrian corridors throughout the central city.

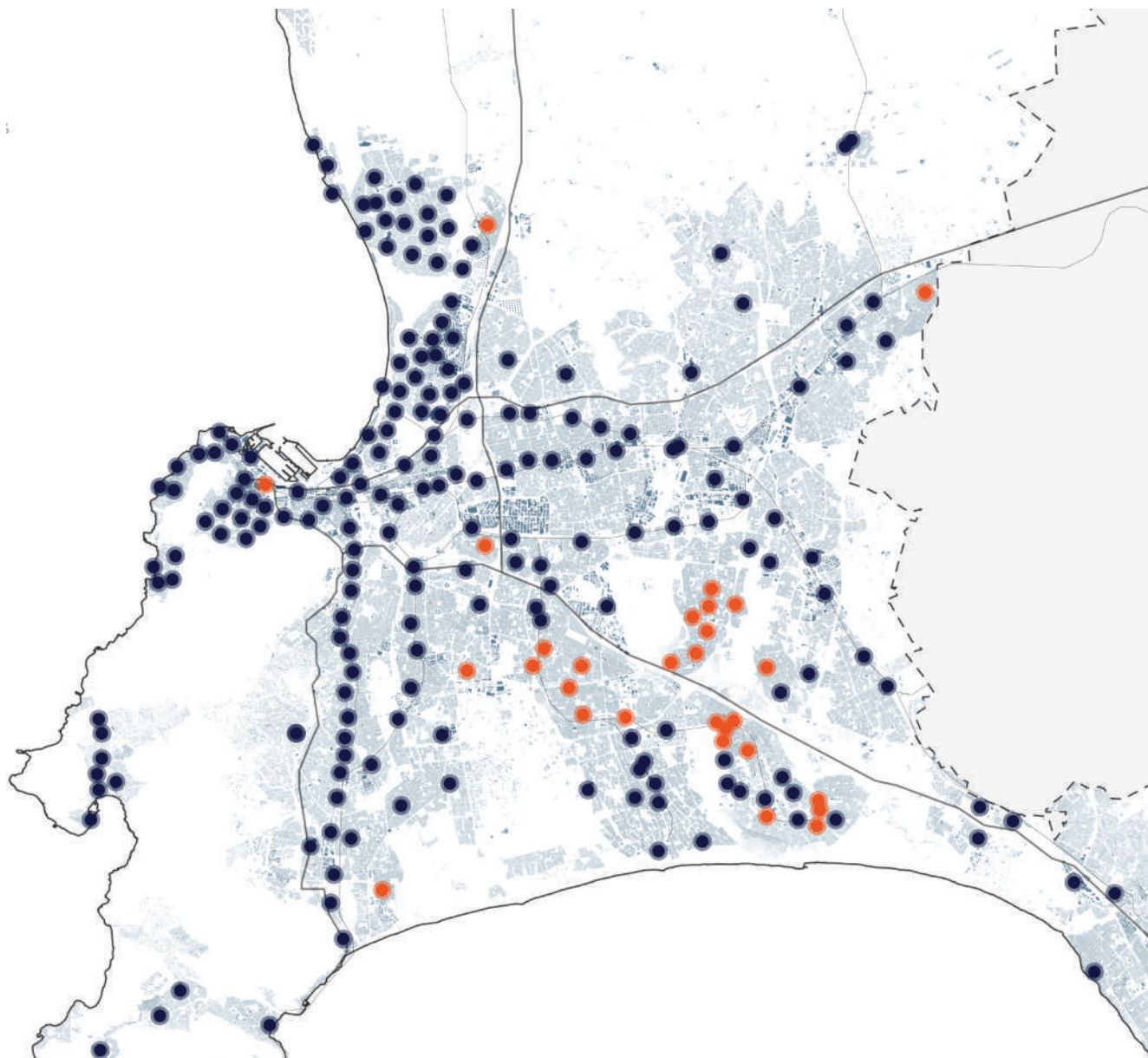
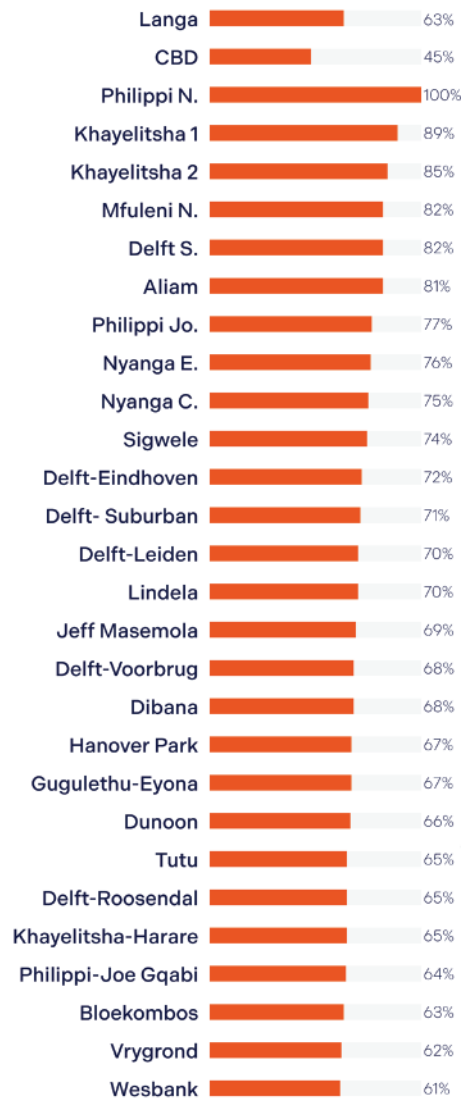


Strand Street, CBD, Opportunity Hub Intervention 2050
Source © [Cape Town] Cohort, Norman Foster Institute. 2025

Comparative Insights

The two pilot sites demonstrate the methodology's capacity to generate contextually appropriate responses from standardised analytical and design tools. Langa's intervention emphasises the needs of residential communities through larger-scale pavilions with longer dwell-time programming, while the CBD intervention responds to transient populations through compact, quick-service elements. Nevertheless, both derive from the same analytical framework and draw from the same catalogue matrix, adapted through KPI weighting and community input to local conditions.

These pilots establish the foundation for systematic expansion across Cape Town's transport network. The lessons derived from Langa and CBD implementation would inform deployment at subsequent priority nodes identified through the Opportunity Explorer. The 2026 pilot phase is a targeted approach, and successful demonstrations position the intervention for scaling, one pilot at a time, progressively transforming transport nodes from spaces of waiting into spaces of opportunity, with the aspiration that these initial demonstrations will catalyse a city-wide movement toward spatial justice.



Thirty nodes in the highest need of Opportunity Hubs
 Source © [Cape Town] Cohort, Norman Foster Institute. 2025

8. Conclusions



Vision of Opportunity Hubs for Cape Town

Source © [Cape Town] Cohort, Norman Foster Institute. 2025

Cape Town's spatial inequality demands intervention that matches the scale and complexity of the challenge. This report presents the Opportunity Hub Network as a systematic response: transforming areas near and around transport interchanges from spaces of waiting into platforms for opportunity, meeting people where they already are, rather than requiring additional travel to bring them elsewhere.

The report's primary contribution is methodological. The five-stage framework bridges the gap between analysis and implementation. Data informs but does not dictate design, and standardisation enables rather than constrains local expression. The Opportunity Explorer tool ensures investment responds to demonstrated need across 610 transport nodes. Site selection embeds services within existing movement patterns. The catalogue and modular construction systems enable rapid, flexible deployment responsive to both site constraints and community priorities. Together, these components address the compounding deficits in basic services, learning, work, digital access, and civic participation that define spatial disadvantage in Cape Town.

Pilot applications in Langa and the CBD show how this methodology generates contextually appropriate responses. Langa's larger pavilions serve residential communities, while CBD interventions respond to transient commuters through compact, quick-service elements.

These pilots are designed to deliver improved access to essential services, skills development, and economic

opportunity at locations where hundreds of thousands of residents already pass daily, converting unproductive waiting time into pathways for advancement.

Strong alignment with Cape Town's policy frameworks creates favourable conditions for adoption. The Integrated Development Plan's commitments to spatial integration, the Comprehensive Integrated Transport Plan's triple access paradigm, and the recent Constitutional Court's recognition of spatial apartheid's ongoing harms all support the intervention's logic and provide a strong foundation for implementation.

This research acknowledges its limitations. The analysis relies on municipal and publicly available data rather than primary collection. And a single field visit limited opportunities for deeper community engagement. Pilots should be understood as demonstrations requiring stronger local partnerships and buy-in before implementation.

Despite these constraints, the research establishes a foundation for action. The next phase requires securing institutional champions within the City and with external partners for Cape Town, identifying financing mechanisms, and committing to initial pilots to test assumptions and generate learning. Cape Town's spatial form was deliberately constructed through segregationist planning. Its transformation requires equally deliberate intervention. The Opportunity Hub Network offers a methodology for beginning that work, toward a city where access to opportunity no longer depends on where one lives.

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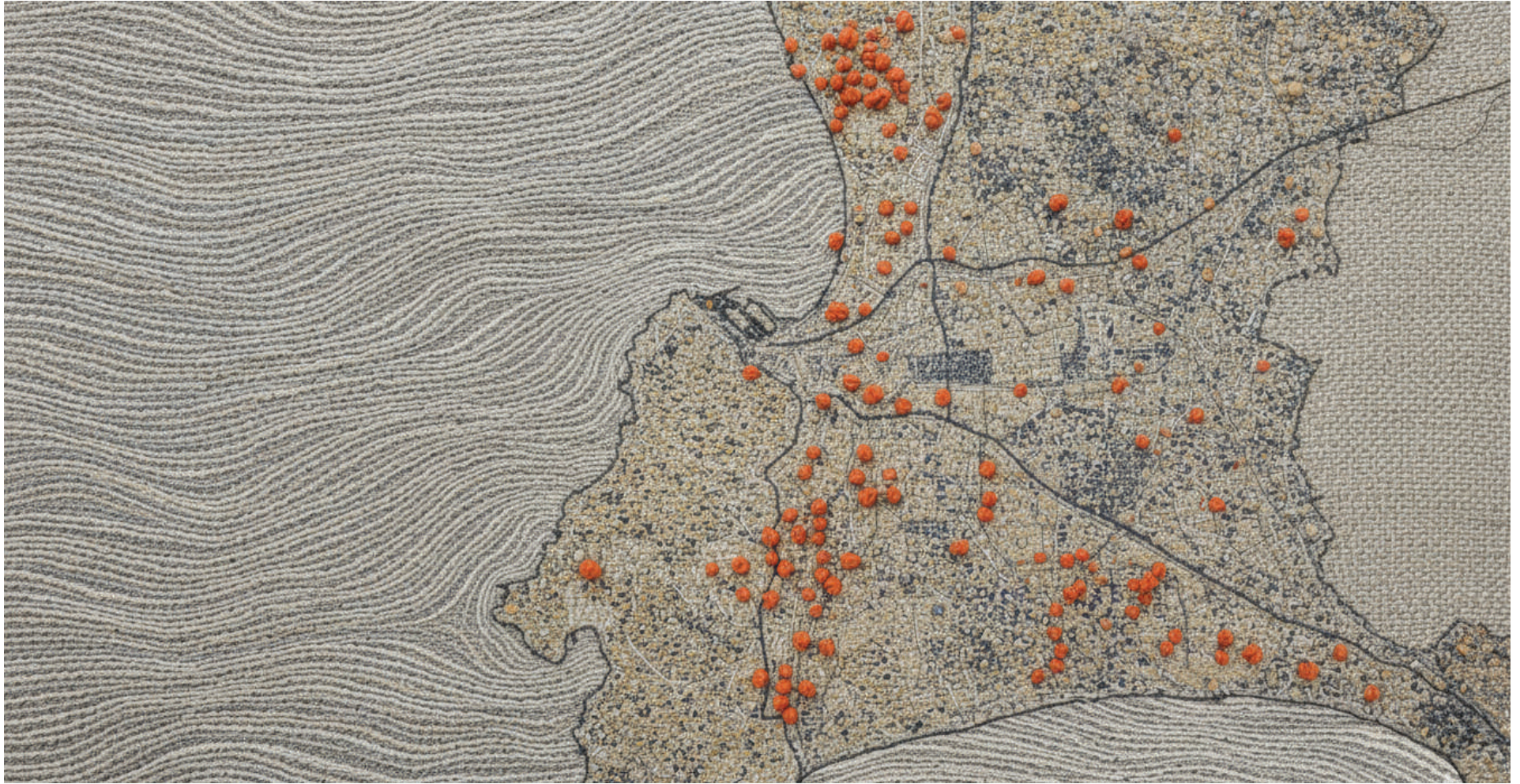
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Vision manifested in a Tapestry

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Team Photo on Top of Table Mountain

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